

Intel® Xeon® Processor E5-2400 Product Family

Thermal / Mechanical Design Guide

May 2012



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Contents

1	Introduction	9
1.1	References	10
1.2	Definition of Terms	10
2	LGA1356 Socket	13
2.1	Board Layout	15
2.2	Attachment to Motherboard	16
2.3	Socket Components	16
2.3.1	Socket Body Housing	16
2.3.2	Solder Balls	16
2.3.3	Contacts	17
2.3.4	Pick and Place Cover	17
2.4	Package Installation / Removal	18
2.4.1	Socket Standoffs and Package Seating Plane	19
2.5	Durability	19
2.6	Markings	19
2.7	Component Insertion Forces	20
2.8	Socket Size	20
2.9	LGA1356 Socket NCTF Solder Joints	20
3	Independent Loading Mechanism (ILM) and Back Plate	23
3.1	Design Concept	23
3.1.1	ILM Assembly Design Overview	23
3.1.2	ILM Back Plate Design Overview	24
3.1.3	Durability	24
3.2	Assembly of ILM to a Motherboard	25
3.3	ILM Cover	27
4	LGA1356 Socket, ILM and Back Plate Electrical, Mechanical, and Environmental Specifications	29
4.1	Component Mass	29
4.2	Package/Socket Stackup Height	29
4.3	Socket Maximum Temperature	29
4.4	Loading Specifications	30
4.5	Electrical Requirements	30
4.6	Environmental Requirements	31
5	Thermal Solutions	33
5.1	Boundary Conditions	33
5.2	Assembly	35
5.2.1	Thermal Interface Material (TIM)	36
5.3	Structural Considerations	36
5.4	Thermal Design	36
5.4.1	Thermal Characterization Parameter	36
5.5	Fan Speed Control	37
5.5.1	Fundamentals	37
5.6	Thermal Features	37
5.6.1	TCONTROL and DTS Relationship	38
5.6.2	Short Duration TCC Activation and Catastrophic Thermal Management for Intel® Xeon® Processor E5-2400 Product Family	39
5.6.3	Intel® Turbo Boost Technology	40
5.7	Thermal Guidance	40
5.7.1	Thermal Excursion	40
5.7.2	Absolute Processor Temperature	40



5.8	DTS Based Thermal Specification	41
5.8.1	Compliance to Tcase Based Thermal Profile	41
5.8.2	Considerations for Follow-on Processor	41
5.8.3	DTS Based Thermal Profile, Tcontrol and Margin for the Intel® Xeon® Processor E5-2400 Product Family	41
5.8.4	Power Calculation for the Intel® Xeon® Processor E5-2400 Product Family	42
5.8.5	Averaging the DTS Based Thermal Specification for the Intel® Xeon® Processor E5-2400 Product Family	42
5.8.6	Capabilities for the Follow-on Processor	43
6	Quality and Reliability Requirements	45
6.1	Test Conditions	45
6.2	Intel Reference Component Validation	45
6.2.1	Board Functional Test Sequence	45
6.2.2	Post-Test Pass Criteria	45
6.2.3	Recommended BIOS/Processor/Memory Test Procedures	46
6.3	Material and Recycling Requirements	46
A	Component Suppliers	47
A.1	Intel Enabled Supplier Information	47
A.1.1	Intel Reference Thermal Solution	47
A.1.2	Intel Collaboration Thermal Solution	47
A.1.3	Alternative Thermal Solution	48
A.1.4	Socket, ILM and Back Plate	50
B	Mechanical Drawings	51
C	Socket Mechanical Drawings	85
D	Processor Installation Tool	91
E	Embedded Thermal Solutions	93
E.1	Performance Targets	93
E.2	Thermal Design Guidelines	94
E.2.1	High Case Temperature Thermal Profile	94
E.3	Mechanical Drawings and Supplier Information	95

Figures

1-1	Intel® Xeon® Processor E5-2400 Product Family Platform Socket Stack	9
2-1	LGA1356 Socket with Pick and Place Cover Removed	13
2-2	LGA1356 Socket Contact Numbering (Top View of Socket)	14
2-3	LGA1356 Socket Land Pattern (Top View of Board)	15
2-4	Attachment to Motherboard	16
2-5	Pick and Place Cover	17
2-6	Package Installation / Removal Features	18
2-7	Package and Board Enabling Mark (-2)	19
2-8	LGA1356 NCTF Solder Joints	21
3-1	ILM Assembly	24
3-2	Back Plate	25
3-3	ILM Assembly	26
3-4	Pin1 and ILM Lever	27
4-1	Flow Chart of Knowledge-Based Reliability Evaluation Methodology	32
5-1	Best-fit Equations	34
5-2	1U Reference Heatsink Assembly	35
5-3	Processor Thermal Characterization Parameter Relationships	37
B-1	Board Keepin / Keepout Zones (Sheet 1 of 4)	52



B-2	Board Keepin / Keepout Zones (Sheet 2 of 4).....	53
B-3	Board Keepin / Keepout Zones (Sheet 3 of 4).....	54
B-4	Board Keepin / Keepout Zones (Sheet 4 of 4).....	55
B-5	1U Reference Heatsink Assembly (Sheet 1 of 2)	56
B-6	1U Reference Heatsink Assembly (Sheet 2 of 2)	57
B-7	1U Reference Heatsink Fin and Base (Sheet 1 of 2).....	58
B-8	1U Reference Heatsink Fin and Base (Sheet 2 of 2).....	59
B-9	Heatsink Shoulder Screw (1U, 2U and Tower)	60
B-10	Heatsink Compression Spring (1U, 2U and Tower)	61
B-11	Heatsink Retaining Ring (1U, 2U and Tower)	62
B-12	Heatsink Load Cup (1U, 2U and Tower)	63
B-13	2U Collaborative Heatsink Assembly (Sheet 1 of 2)	64
B-14	2U Collaborative Heatsink Assembly (Sheet 2 of 2)	65
B-15	2U Collaborative Heatsink Volumetric (Sheet 1 of 2)	66
B-16	2U Collaborative Heatsink Volumetric (Sheet 2 of 2)	67
B-17	Tower Collaborative Heatsink Assembly (Sheet 1 of 2)	68
B-18	Tower Collaborative Heatsink Assembly (Sheet 2 of 2)	69
B-19	Tower Collaborative Heatsink Volumetric (Sheet 1 of 2).....	70
B-20	Tower Collaborative Heatsink Volumetric (Sheet 2 of 2).....	71
B-21	1U Reference Heatsink Assembly with TIM (Sheet 1 of 2)	72
B-22	1U Reference Heatsink Assembly with TIM (Sheet 2 of 2)	73
B-23	2U Reference Heatsink Assembly with TIM (Sheet 1 of 2)	74
B-24	2U Reference Heatsink Assembly with TIM (Sheet 2 of 2)	75
B-25	Tower Reference Heatsink Assembly with TIM (Sheet 1 of 2).....	76
B-26	Tower Reference Heatsink Assembly with TIM (Sheet 2 of 2).....	77
B-27	25.5 mm Reference Heatsink Assembly (Sheet 1 of 2)	78
B-28	25.5 mm Reference Heatsink Assembly (Sheet 2 of 2)	79
B-29	25.5 mm Reference Heatsink Fin and Base (Sheet 1 of 2).....	80
B-30	25.5 mm Reference Heatsink Fin and Base (Sheet 2 of 2).....	81
B-31	25.5 mm Reference Heatsink Assembly with TIM (Sheet 1 of 2).....	82
B-32	25.5 mm Reference Heatsink Assembly with TIM (Sheet 2 of 2).....	83
C-1	Socket Mechanical Drawing (Sheet 1 of 4)	86
C-2	Socket Mechanical Drawing (Sheet 2 of 4)	87
C-3	Socket Mechanical Drawing (Sheet 3 of 4)	88
C-4	Socket Mechanical Drawing (Sheet 4 of 4)	89
D-1	Processor Installation Tool	92
E-1	ATCA Heatsink Performance Curves	94
E-2	NEBS Thermal Profile	95
E-3	ATCA Reference Heat Sink Assembly (Sheet 1 of 2)	97
E-4	ATCA Reference Heat Sink Assembly (Sheet 2 of 2)	98
E-5	ATCA Reference Heatsink Fin and Base (Sheet 1 of 2)	99
E-6	ATCA Reference Heatsink Fin and Base (Sheet 2 of 2)	100

Tables

1-1	Reference Documents	10
1-2	Terms and Descriptions.....	10
4-1	Component Mass.....	29
4-2	1356-land Package and LGA1356 Socket Stackup Height	29
4-3	Socket and ILM Mechanical Specifications	30
4-4	Electrical Requirements for LGA1356 Socket	31
5-1	Values Used to Generate Processor Thermal Specifications.....	33
5-2	Performance Expectations in Compact Electronics Bay (CEB)	34



5-3	TCONTROL and DTS Relationship	38
5-4	Sign Convention	38
5-5	T _{CONTROL} Relief for Intel® Xeon® Processor E5-2400 Product Family	39
5-6	Averaging Coefficients	43
A-1	Suppliers for the Intel Reference Thermal Solution	47
A-2	Suppliers for the Intel Collaboration Thermal Solution	48
A-3	Suppliers for the Alternative Thermal Solution	48
A-4	LGA1356 Socket, ILM and Back Plate	50
B-1	Mechanical Drawing List	51
C-1	Mechanical Drawing List	85
E-1	8-Core/6-Core Processor Reference Thermal Boundary Conditions	93
E-2	4-Core Processor Reference Thermal Boundary Conditions	93
E-3	Embedded Heatsink Component Suppliers	95
E-4	Mechanical Drawings List	96



Revision History

Document Number	Revision Number	Description	Date
327250	-001	Initial release of the document.	May 2012

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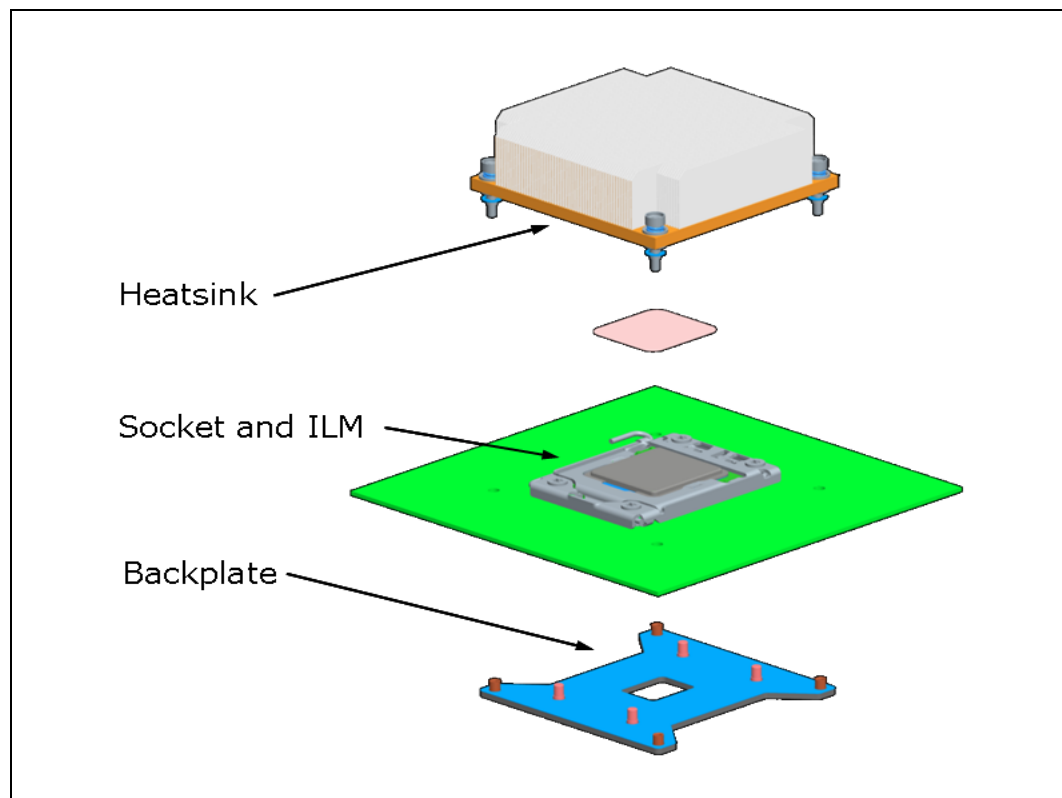


1 Introduction

This document provides guidelines for the design of thermal and mechanical solutions for server and workstation processors in the Intel® Xeon® Processor E5-2400 Product Family platform. The processors covered include those listed in the *Intel® Xeon® Processor E5-2400 Product Family Datasheet - Volume One*. The components described in this document include:

- The processor thermal solution (heatsink) and associated retention hardware.
- The LGA1356 socket, the Independent Loading Mechanism (ILM) and back plate.

Figure 1-1. Intel® Xeon® Processor E5-2400 Product Family Platform Socket Stack



The goals of this document are:

- To assist board and system thermal mechanical designers.
- To assist designers and suppliers of processor heatsinks.

Thermal profiles and other processor specifications are provided in the appropriate Datasheet.



1.1 References

Material and concepts available in the following documents may be beneficial when reading this document.

Table 1-1. Reference Documents

Document	Number	Notes
European Blue Angel Recycling Standards		2
Intel® Xeon® Processor E5-2400 Product Family Datasheet - Volume One	327248	1
Platform Environment Control Interface (PECI) Specification		4
Intel® Xeon® Processor E5-2400 Processor Product Family Mechanical Model	327322	1
Intel® Xeon® Processor E5-2400 Processor Product Family Thermal Model	327321	1
Manufacturing With Intel Components Using Lead-Free Technology		3
Platform Digital Thermal Sensor (DTS) Based Thermal Specifications and Overview		4

Notes:

1. Available at <http://www.intel.com>. Document numbers are subject to change.
2. Available at <http://www.blauer-engel.de/en/index.php>
3. Available at <https://learn.intel.com/portal/scripts/general/logon.aspx>.
4. Contact your local Intel Field Sales Representative.

1.2 Definition of Terms

Table 1-2. Terms and Descriptions (Sheet 1 of 2)

Term	Description
Bypass	Bypass is the area between a passive heatsink and any object that can act to form a duct. For this example, it can be expressed as a dimension away from the outside dimension of the fins to the nearest surface.
DTS	Digital Thermal Sensor reports a relative die temperature as an offset from TCC activation temperature.
FSC	Fan Speed Control
IHS	Integrated Heat Spreader: a component of the processor package used to enhance the thermal performance of the package. Component thermal solutions interface with the processor at the IHS surface.
ILM	Independent Loading Mechanism provides the force needed to seat the 1356-LGA land package onto the socket contacts.
LGA1356 socket	The processor mates with the system board through this surface mount, 1356-contact socket.
PECI	The Platform Environment Control Interface (PECI) is a one-wire interface that provides a communication channel between Intel processor and chipset components to external monitoring devices.
Ψ_{CA}	Case-to-ambient thermal characterization parameter (psi). A measure of thermal solution performance using total package power. Defined as $(T_{CASE} - T_{LA}) / \text{Total Package Power}$. Heat source should always be specified for Ψ measurements.
Ψ_{CS}	Case-to-sink thermal characterization parameter. A measure of thermal interface material performance using total package power. Defined as $(T_{CASE} - T_S) / \text{Total Package Power}$.
Ψ_{SA}	Sink-to-ambient thermal characterization parameter. A measure of heatsink thermal performance using total package power. Defined as $(T_S - T_{LA}) / \text{Total Package Power}$.



Table 1-2. Terms and Descriptions (Sheet 2 of 2)

Term	Description
T_{CASE}	The case temperature of the processor measured at the geometric center of the topside of the IHS.
T_{CASE_MAX}	The maximum case temperature as specified in a component specification.
TCC	Thermal Control Circuit: Thermal monitor uses the TCC to reduce the die temperature by using clock modulation and/or operating frequency and input voltage adjustment when the die temperature is very near its operating limits.
$T_{CONTROL}$	$T_{CONTROL}$ is a static value below TCC activation used as a trigger point for fan speed control.
TDP	Thermal Design Power: Thermal solution should be designed to dissipate this target power level. TDP is not the maximum power that the processor can dissipate.
Thermal Monitor	A power reduction feature designed to decrease temperature after the processor has reached its maximum operating temperature.
Thermal Profile	Line that defines the temperature specification of a processor at a given power level.
TIM	Thermal Interface Material: The thermally conductive compound between the heatsink and the processor case. This material fills the air gaps and voids, and enhances the transfer of the heat from the processor case to the heatsink.
T_{LA}	The measured ambient temperature locally surrounding the processor. The ambient temperature should be measured just upstream of a passive heatsink or at the fan inlet for an active heatsink.
T_{SA}	The system ambient air temperature external to a system chassis. This temperature is usually measured at the chassis air inlets.
U	A unit of measure used to define server rack spacing height. 1U is equal to 1.75 in, 2U equals 3.50 in, etc.

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2 LGA1356 Socket

This chapter describes a surface mount, LGA (Land Grid Array) socket intended for processors in the E5-2400 Product Family Platform. The socket provides I/O, power and ground contacts. The socket contains 1356 contacts arrayed about a cavity in the center of the socket with lead-free solder balls for surface mounting on the motherboard.

The socket has 1356 contacts with 1.016 mm X 1.016 mm pitch (X by Y) in a 43x41 grid array with 21x17 grid depopulation in the center of the array and selective depopulation elsewhere.

The socket must be compatible with the package (processor) and the Independent Loading Mechanism (ILM). The design includes a back plate which is a key contributor in producing a uniform load on the socket solder joints. Socket loading specifications are listed in [Section 4.4](#).

Figure 2-1. LGA1356 Socket with Pick and Place Cover Removed

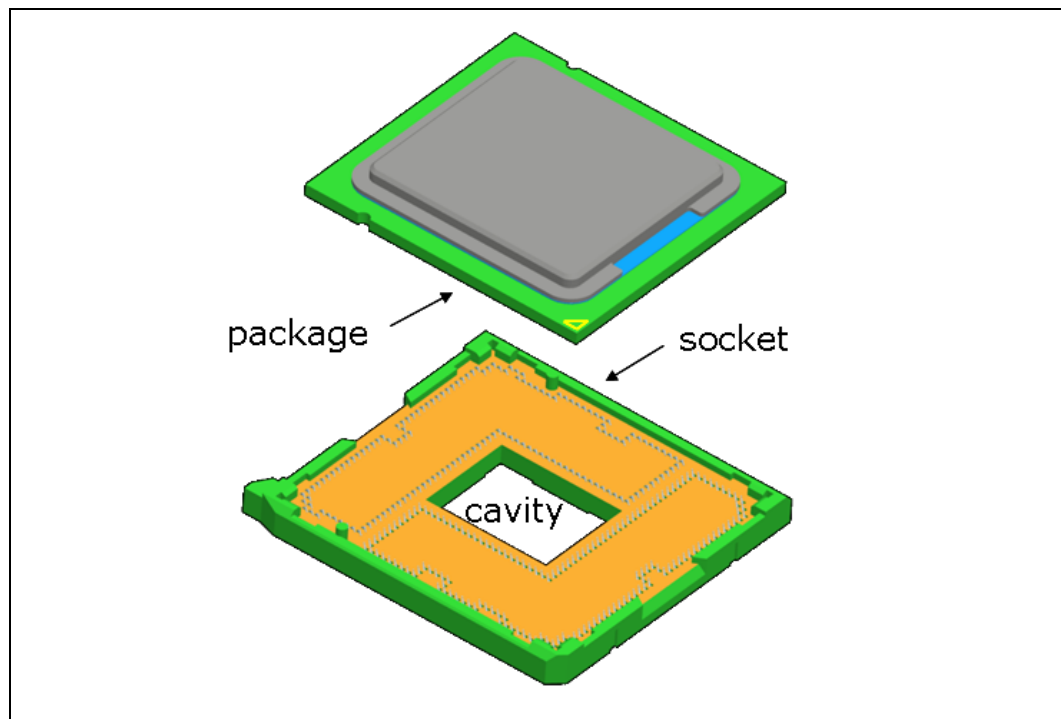
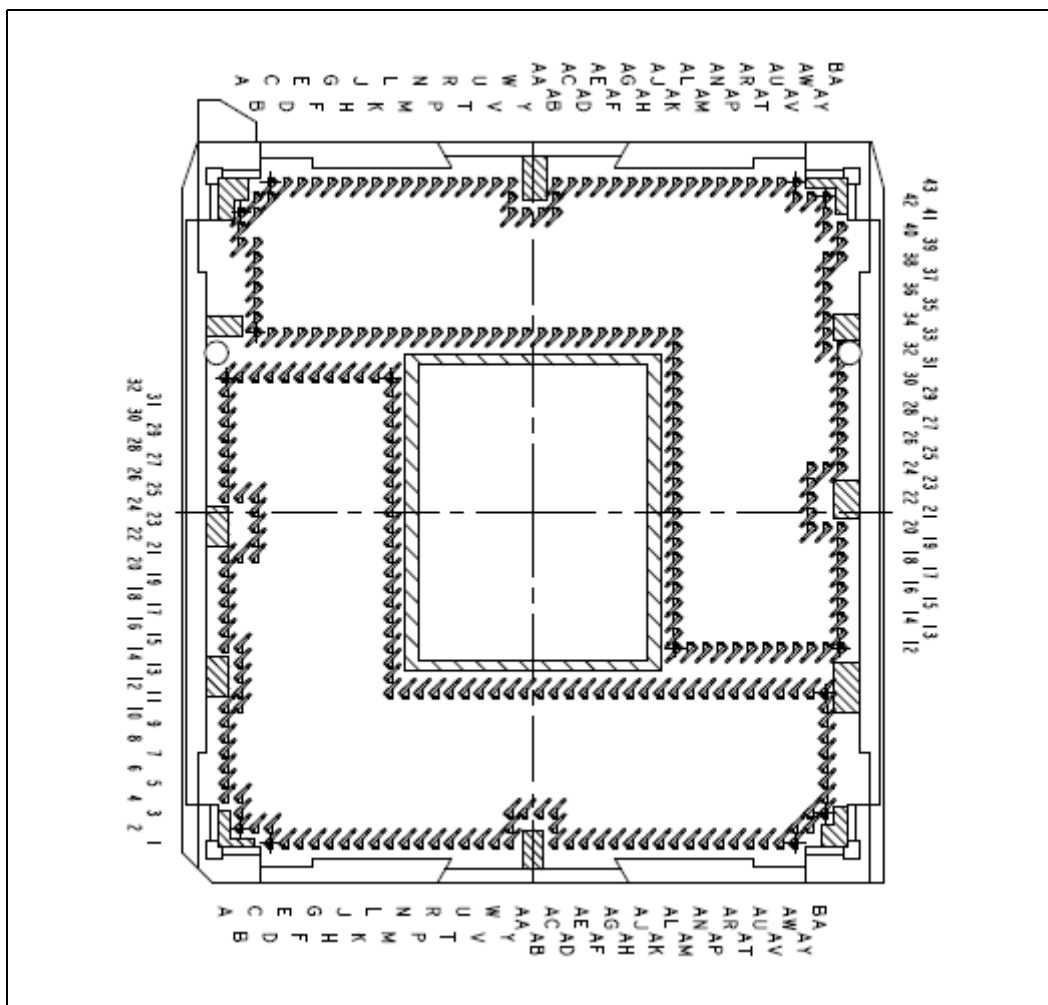


Figure 2-2. LGA1356 Socket Contact Numbering (Top View of Socket)

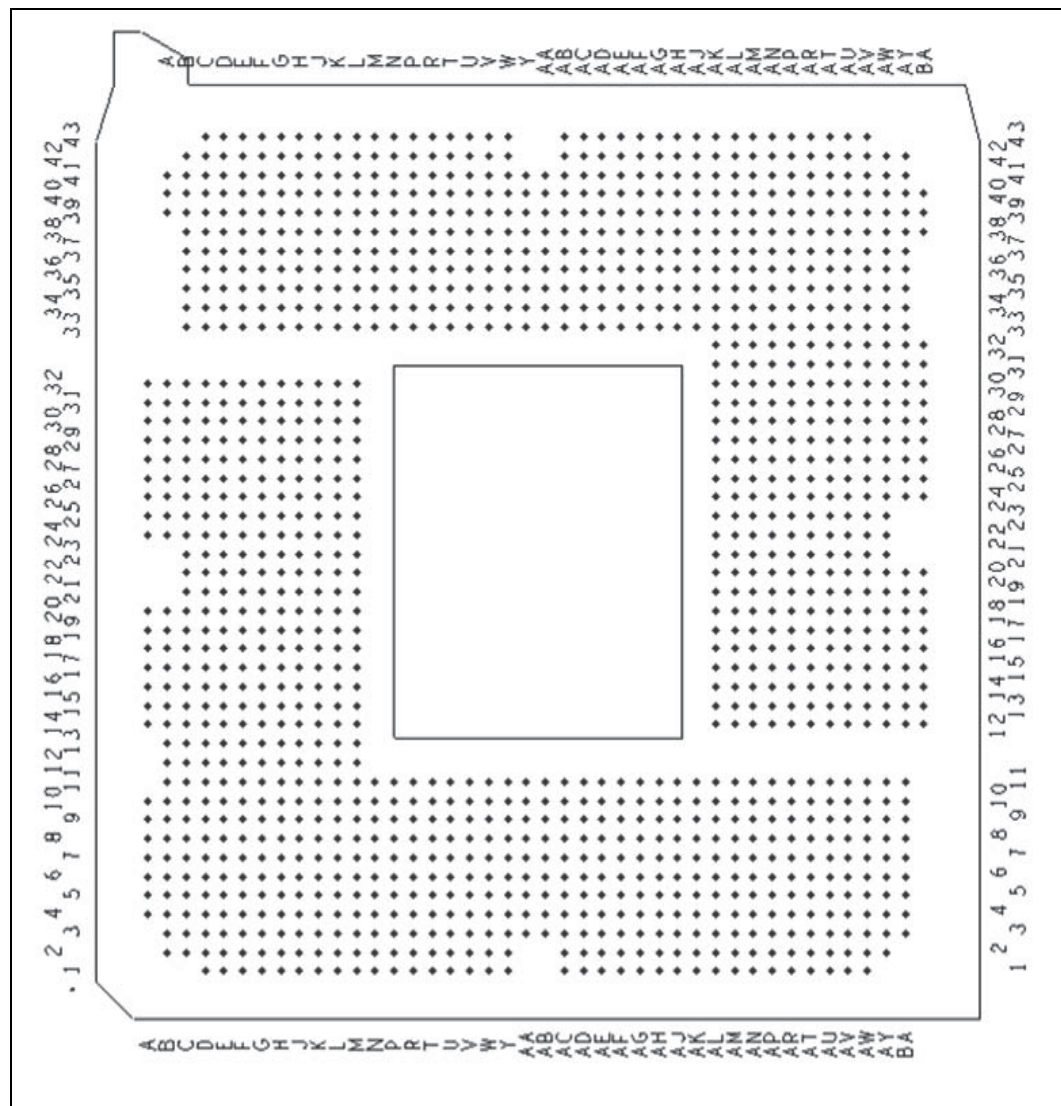


2.1 Board Layout

The land pattern for the LGA1356 socket is 40 mils X 40 mils (X by Y). Note that there is no round-off (conversion) error between socket pitch (1.016 mm) and board pitch (40 mil) as these values are equivalent.

In general, metal defined (MD) pads perform better than solder mask defined (SMD) pads under thermal cycling, and SMD pads perform better than MD pads under dynamic stress. At this time, complete recommendations for pad definition and pad size do not exist for the LGA1356 socket. See [Section 2.9](#) for more information on pad definition and pad size.

Figure 2-3. LGA1356 Socket Land Pattern (Top View of Board)

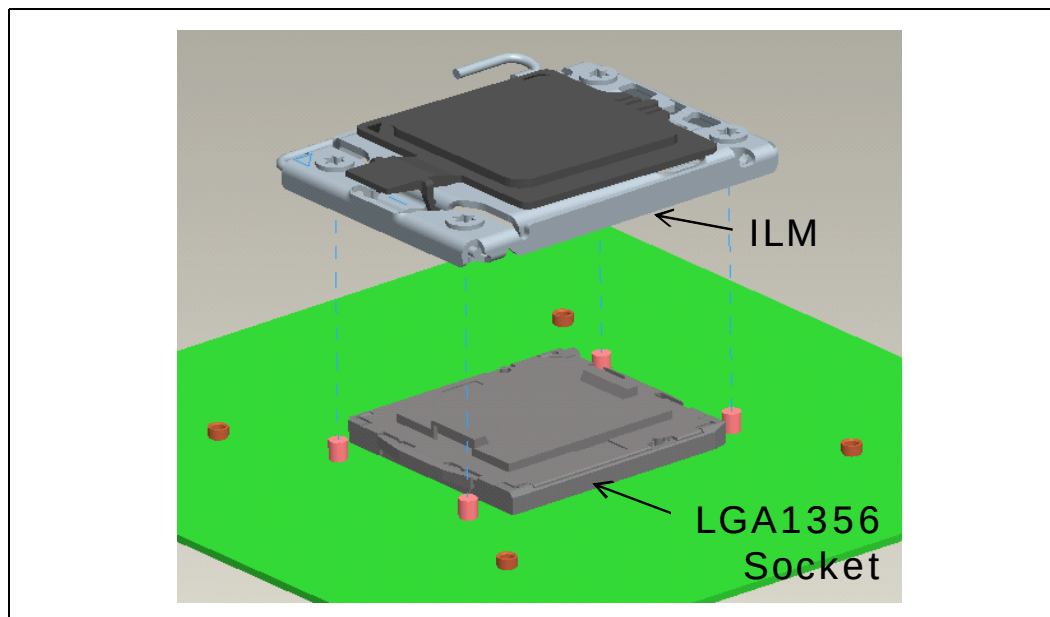


2.2 Attachment to Motherboard

The socket is attached to the motherboard by 1356 solder balls. There are no additional external methods (that is, screw, extra solder, adhesive, and so on) to attach the socket.

As indicated in [Figure 2-4](#), the Independent Loading Mechanism (ILM) is not present during the attach (reflow) process.

Figure 2-4. Attachment to Motherboard



2.3 Socket Components

The socket has two main components, the socket body and Pick and Place (PnP) cover, and is delivered as a single integral assembly. Refer to [Appendix C](#) for detailed drawings.

2.3.1 Socket Body Housing

The housing material is thermoplastic or equivalent with UL 94 V-0 flame rating capable of withstanding 260 °C for 40 seconds (typical reflow/rework). The socket coefficient of thermal expansion (in the XY plane), and creep properties, must be such that the integrity of the socket is maintained for the conditions listed in the LGA1366 Socket Validation Reports, and the LGA1356 Addendum.

The color of the housing will be dark as compared to the solder balls to provide the contrast needed for pick and place vision systems.

2.3.2 Solder Balls

A total of 1356 solder balls corresponding to the contacts are on the bottom of the socket for surface mounting with the motherboard.

The socket has the following solder ball material:

- Lead free SAC (SnAgCu) solder alloy with a silver (Ag) content between 3% and 4% and a melting temperature of approximately 217 °C. The alloy must be

compatible with immersion silver (ImAg) motherboard surface finish and a SAC alloy solder paste.

The co-planarity (profile) and true position requirements are defined in [Appendix C](#).

2.3.3 Contacts

Base material for the contacts is high strength copper alloy.

For the area on socket contacts where processor lands will mate, there is a 0.381 μm [15 μinches] minimum gold plating over 1.27 μm [50 μinches] minimum nickel underplate.

No contamination by solder in the contact area is allowed during solder reflow.

2.3.4 Pick and Place Cover

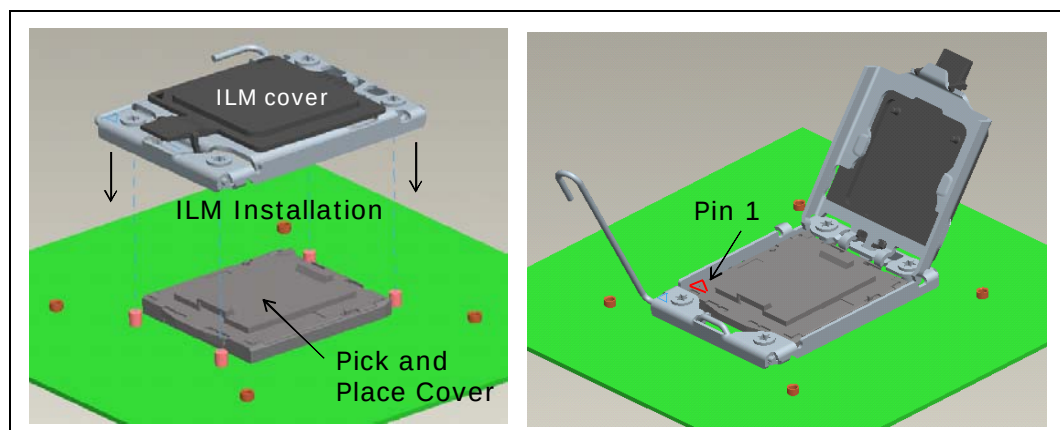
The cover provides a planar surface for vacuum pick up used to place components in the Surface Mount Technology (SMT) manufacturing line. The cover remains on the socket during reflow to help prevent contamination during reflow. The cover can withstand 260 °C for 40 seconds (typical reflow/rework profile) and the conditions listed in the LGA1366 Socket Validation Reports, and LGA1356 Addendum, without degrading. Reports are available from socket suppliers listed in [Appendix A](#).

As indicated in [Figure 2-5](#), the Pick and Place cover remains on the socket during ILM installation. Use of the ILM cover can mitigate against bent socket contacts associated with reinstalling the Pick and Place cover. A cover should remain on whenever possible to help prevent damage to the socket contacts. See [Section 3.2](#) and [Section 3.3](#) for additional information on the ILM cover.

Pick and Place cover retention must be sufficient to support the socket weight during lifting, translation, and placement (board manufacturing), and during board and system shipping and handling.

Pick and Place covers are designed to be interchangeable between socket suppliers. As indicated in [Figure 2-5](#), a Pin1 indicator on the Pick and Place cover provides a visual reference for proper orientation with the socket.

Figure 2-5. Pick and Place Cover



2.4 Package Installation / Removal

As indicated in [Figure 2-6](#), access is provided to facilitate manual installation and removal of the package.

To assist in package orientation and alignment with the socket:

- The package Pin1 triangle and the socket Pin1 chamfer provide visual reference for proper orientation.
- The package substrate has orientation notches along two opposing edges of the package, offset from the centerline. The socket has two corresponding orientation posts to physically prevent mis-orientation of the package. These orientation features also provide initial rough alignment of package to socket.
- As shown in [Figure 2-7](#), the package substrate has a “-2” mark near the orientation notch on the Pin 1 side. Similarly, space has been reserved for a “-2” mark on the motherboard in the Board Keepin / Keepout Zones in [Figure B-1](#) and [Figure B-2](#). These matching marks help prevent system assemblers from installing the incorrect processor into the socket.
- The socket has alignment walls at the four corners to provide final alignment of the package.

See [Appendix D](#) for information regarding a tool designed to provide mechanical assistance during processor installation and removal.

Figure 2-6. Package Installation / Removal Features

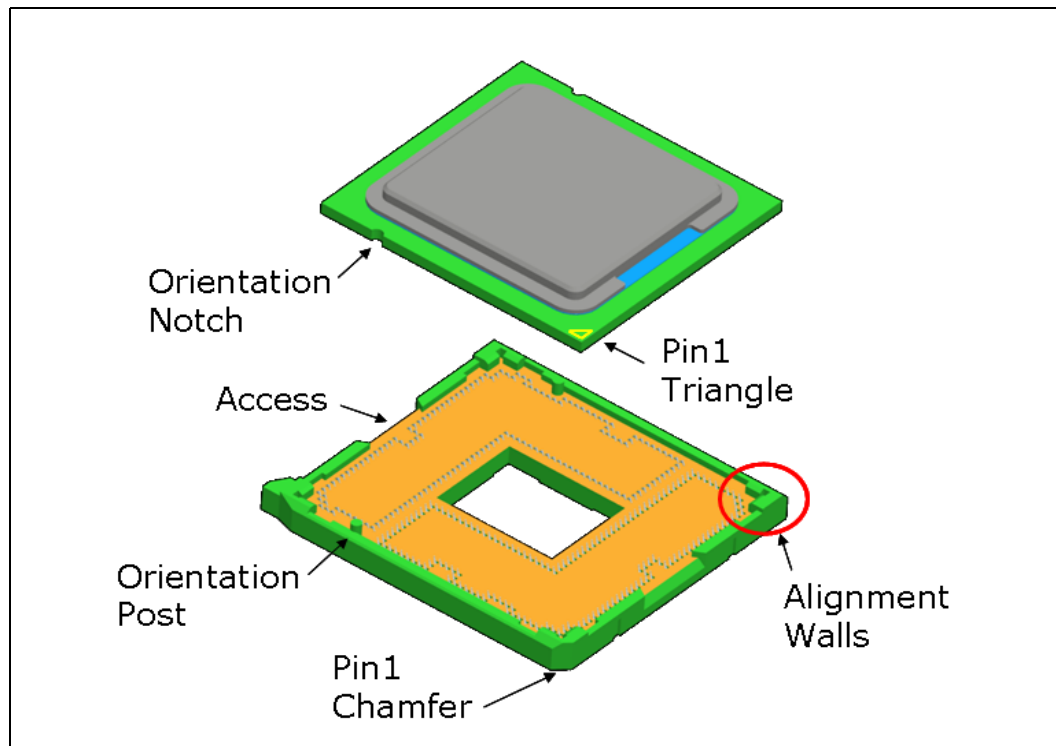
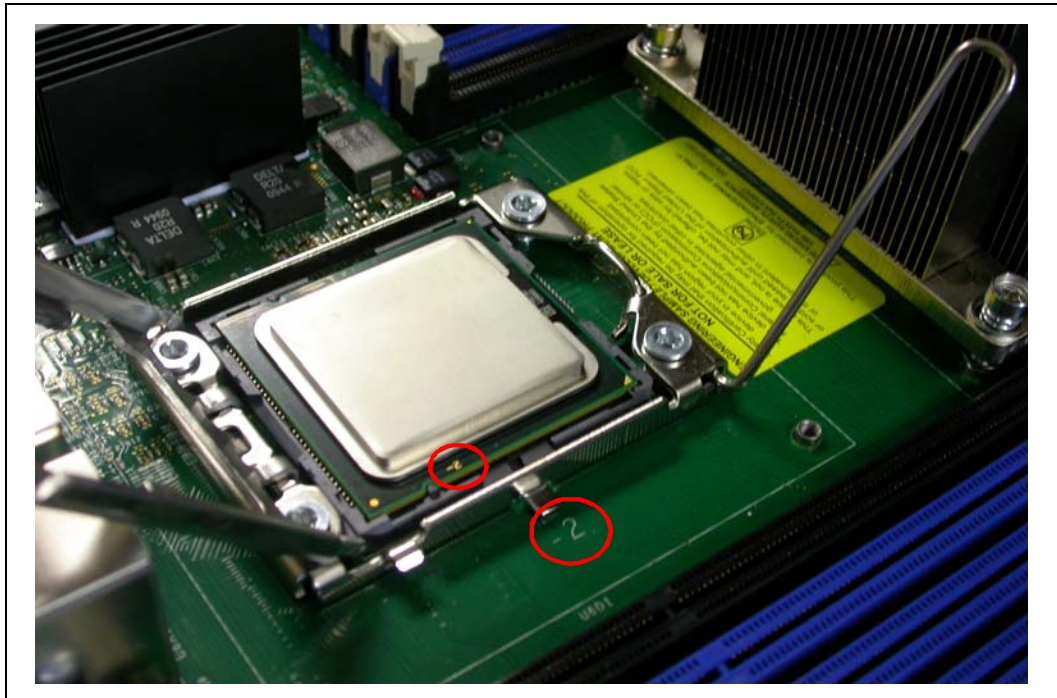


Figure 2-7. Package and Board Enabling Mark (-2)

2.4.1 Socket Standoffs and Package Seating Plane

Standoffs on the bottom of the socket base establish the minimum socket height after solder reflow and are specified in [Appendix C](#).

Similarly, a seating plane on the topside of the socket establishes the minimum package height. See [Section 3.2](#) for the calculated IHS height above the motherboard.

2.5 Durability

The socket must withstand 30 cycles of processor insertion and removal. The max chain contact resistance from [Table 4-4](#) must be met when mated in the 1st and 30th cycles.

The socket Pick and Place cover must withstand 15 cycles of insertion and removal.

2.6 Markings

There are three markings on the socket:

- LGA1356: Font type is Helvetica Bold - minimum 6 point (2.125 mm).
- Manufacturer's insignia (font size at supplier's discretion).
- Lot identification code (allows traceability of manufacturing date and location).

All markings must withstand 260 °C for 40 seconds (typical reflow/rework profile) without degrading, and must be visible after the socket is mounted on the motherboard.

LGA1356 and the manufacturer's insignia are molded or laser marked on the side wall.



2.7 Component Insertion Forces

Any actuation must meet or exceed SEMI S8-95 Safety Guidelines for Ergonomics/ Human Factors Engineering of Semiconductor Manufacturing Equipment, example Table R2-7 (Maximum Grip Forces). The socket must be designed so that it requires no force to insert the package into the socket.

2.8 Socket Size

Socket information needed for motherboard design is given in [Appendix C](#).

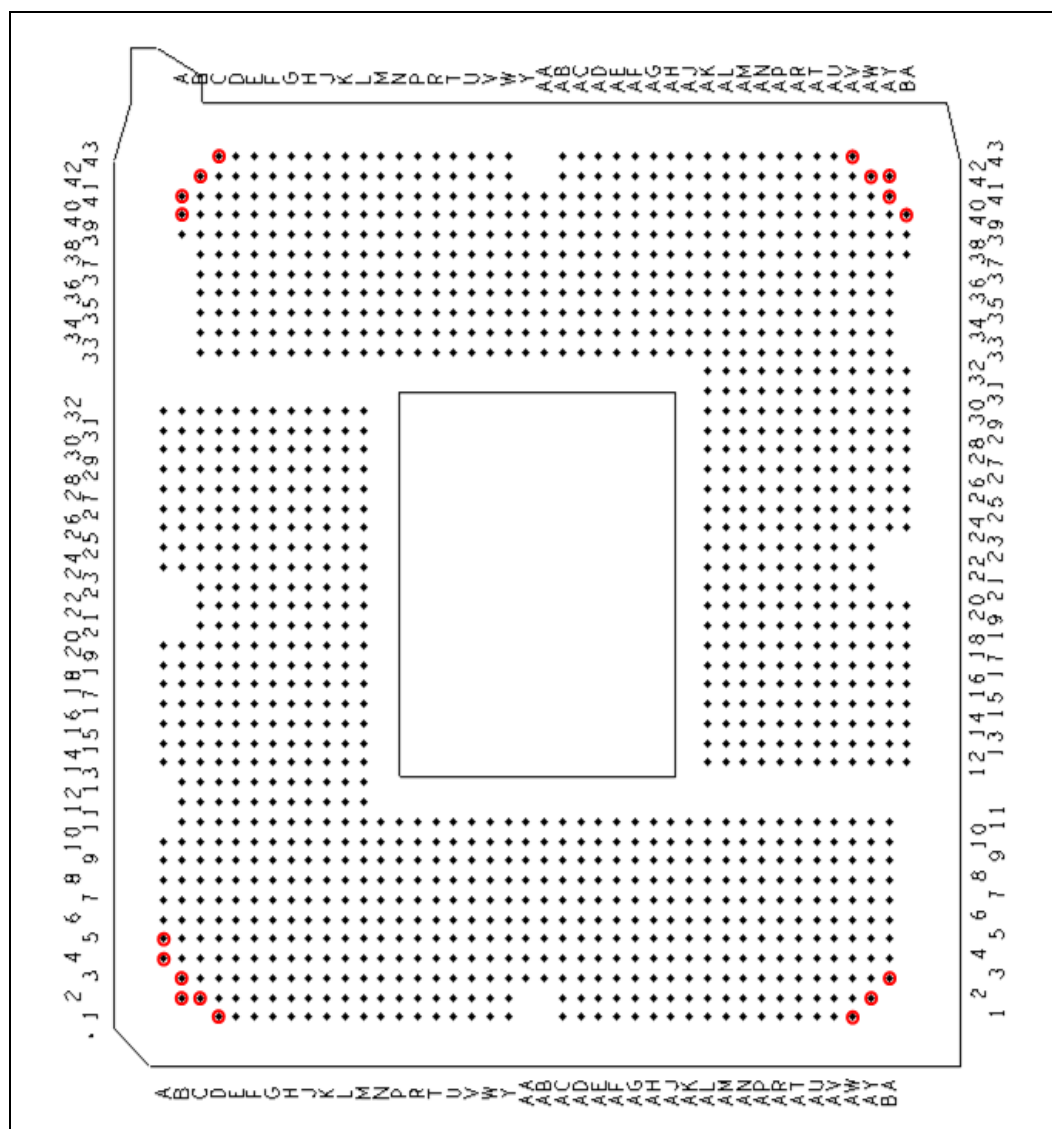
This information should be used in conjunction with the reference motherboard keepout drawings provided in [Appendix B](#) to ensure compatibility with the reference thermal mechanical components.

2.9 LGA1356 Socket NCTF Solder Joints

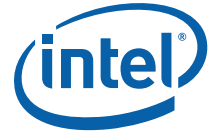
Intel has defined selected solder joints of the socket as non-critical to function (NCTF) for post environmental testing. The processor signals at NCTF locations are typically redundant ground or non-critical reserved, so the loss of the solder joint continuity at end of life conditions will not affect the overall product functionality. [Figure 2-8](#) identifies the NCTF solder joints.

Since corner pads are often more susceptible to solder joint damage, NCTF locations are often placed in the corners. When possible, larger pads may be chosen at NCTF locations to further mitigate against solder joint damage. At this time, complete recommendations for pad definition and pad size do not exist at NCTF locations. CTF and NCTF locations are 18mil solder mask defined on Intel reference designs.

Figure 2-8. LGA1356 NCTF Solder Joints





3 Independent Loading Mechanism (ILM) and Back Plate

The Independent Loading Mechanism (ILM) provides the force needed to seat the 1356-LGA land package onto the socket contacts. The ILM is physically separate from the socket body. The assembly of the ILM to the board is expected to occur after wave solder. The exact assembly location is dependent on manufacturing preference and test flow.

Note: The ILM has two critical functions: deliver the force to seat the processor onto the socket contacts and distribute the resulting compressive load evenly through the socket solder joints.

Note: The mechanical design of the ILM is a key contributor to the overall functionality of the LGA1356 socket. Intel performs detailed studies on integration of processor package, socket and ILM as a system. These studies directly impact the design of the ILM. The Intel reference ILM will be “build to print” from Intel controlled drawings. Intel recommends using the Intel Reference ILM. Custom non-Intel ILM designs do not benefit from Intel's detailed studies and may not incorporate critical design parameters.

3.1 Design Concept

The ILM and back plate are assemblies and can be procured from the enabled vendors.

3.1.1 ILM Assembly Design Overview

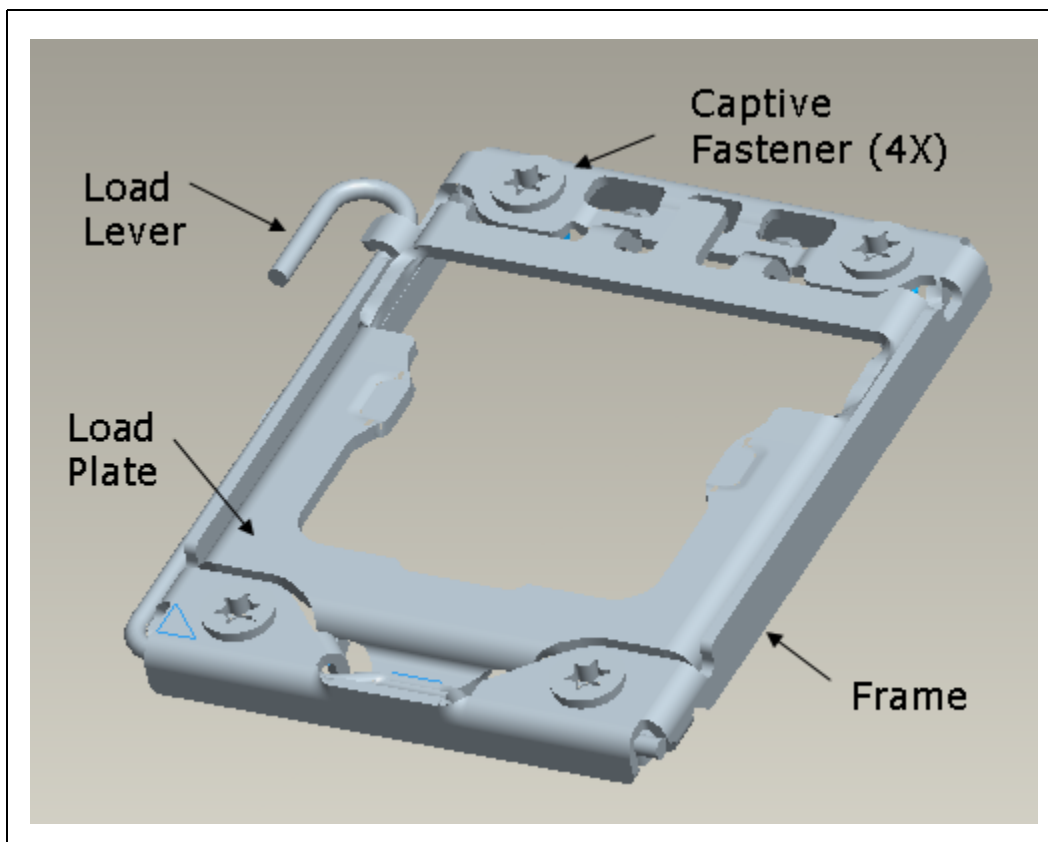
The ILM assembly consists of four major pieces: load lever, load plate, frame and the captive fasteners.

The load lever and load plate are stainless steel. The frame and fasteners are high carbon steel with appropriate plating. The fasteners are fabricated from a high carbon steel. The frame provides the hinge locations for the load lever and load plate.

The ILM assembly design ensures that once assembled to the back plate and the load lever is closed, the only features touching the board are the captive fasteners. The nominal gap of the frame to the board is ~1 mm when the load plate is closed on the empty socket or when closed on the processor package.

When closed, the load plate applies two point loads onto the IHS at the “dimpled” features shown in [Figure 3-1](#). The reaction force from closing the load plate is transmitted to the frame and through the captive fasteners to the back plate. Some of the load is passed through the socket body to the board inducing a slight compression on the solder joints.

Figure 3-1. ILM Assembly



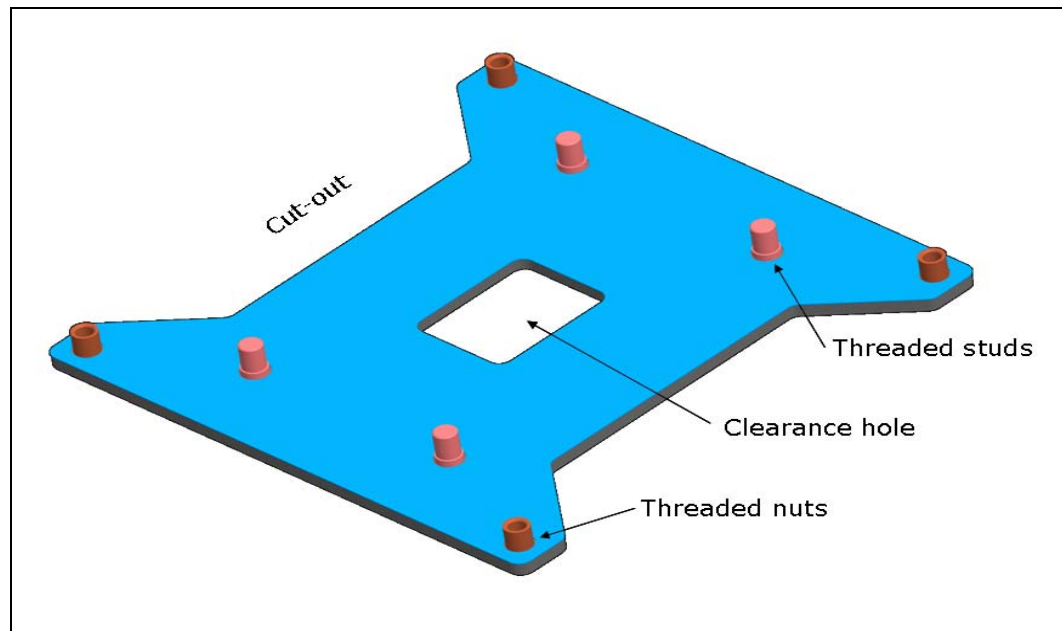
3.1.2 ILM Back Plate Design Overview

The unified back plate consists of a flat steel back plate with threaded studs for ILM attach, and internally threaded nuts for heatsink attach. The threaded studs have a smooth surface feature that provides alignment for the back plate to the motherboard for proper assembly of the ILM around the socket. A clearance hole is located at the center of the plate to allow access to test points and backside capacitors. An additional cut-out on two sides provides clearance for backside voltage regulator components. An insulator is pre-applied. To stay within the temperature limit of the insulator, remove the back plate prior to board component rework.

3.1.3 Durability

The ILM durability requirement is 30 processor cycles. 1 processor cycle = install processor, close load plate, latch load lever, unlatch load lever, open load plate.

The ILM durability requirement is 6 assembly cycles. See [Section 3.2](#) for assembly procedure. 1 assembly cycle = fasten the ILM assembly to the back plate with the four captive screws, torque to 9 ± 1 inch-pounds, unfasten ILM assembly from the back plate.

Figure 3-2. Back Plate

3.2 Assembly of ILM to a Motherboard

The ILM design allows a bottoms up assembly of the components to the board. In step 1 (see [Figure 3-3](#)), the back plate is placed in a fixture. Holes in the motherboard provide alignment to the threaded studs.

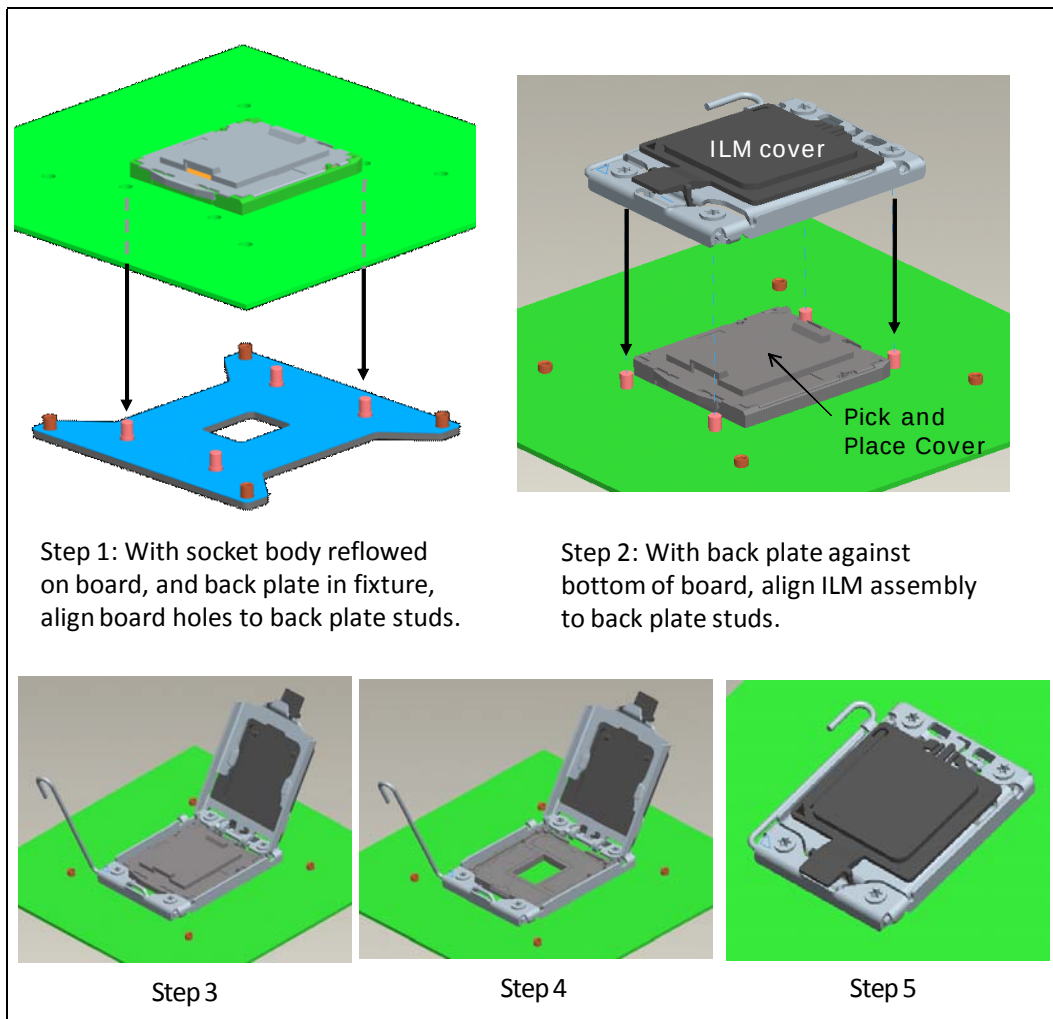
In step 2, the ILM assembly is placed over the socket and threaded studs. The Intel Reference Design ILM cover is not designed to nest over the Pick and Place cover. This feature helps prevent reinstallation of the Pick and Place cover, a step that can lead to socket bent contacts.

To prevent the ILM cover from popping off during ILM assembly, the load plate can be unlatched from the load lever when the fasteners are torqued as shown in Step 3. Using a T20 Torx* driver, fasten the ILM assembly to the back plate with the four captive fasteners. Torque to 9 ± 1 inch-pounds.

The Pick and Place cover can then be removed as shown in Step 4, and the load plate can then be closed and latched as shown in Step 5.

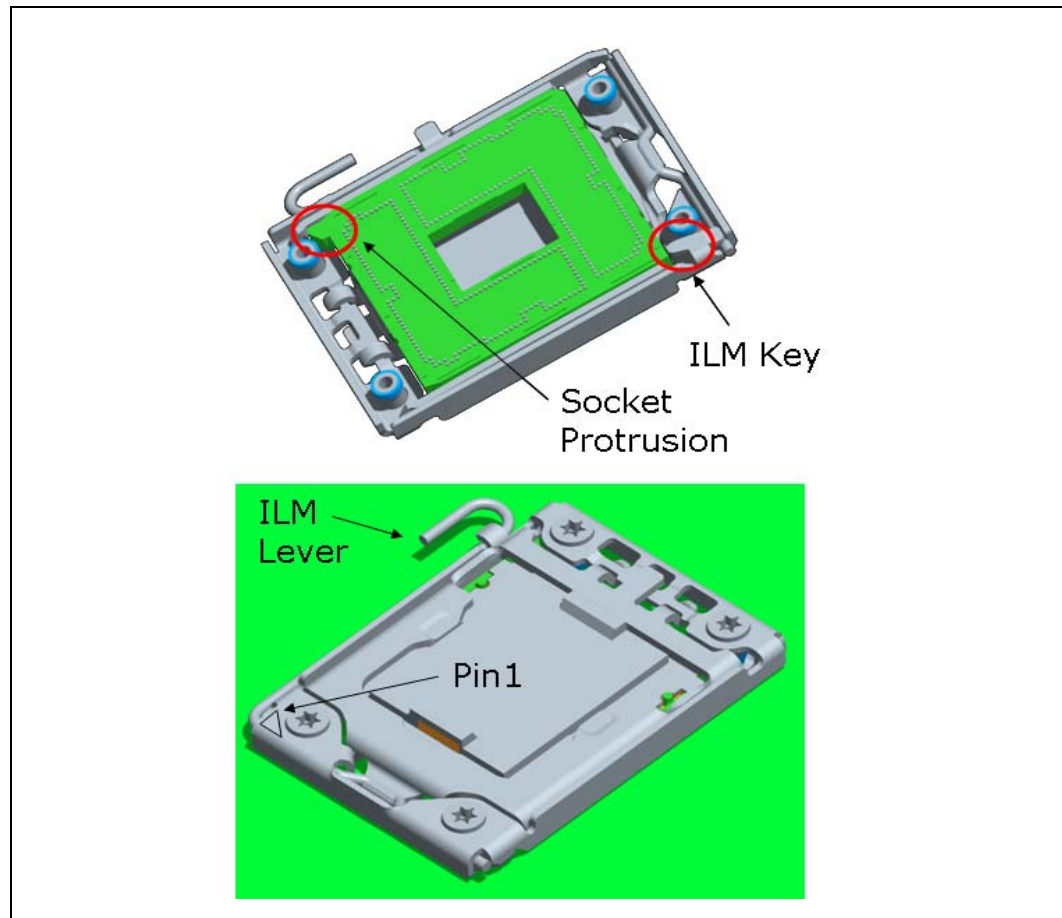
The length of the threaded studs accommodate board thicknesses from 0.062" to 0.100".

Figure 3-3. ILM Assembly



As indicated in [Figure 3-4](#), socket protrusion and ILM key features prevent 180-degree rotation of ILM assembly with respect to the socket. The result is a specific Pin 1 orientation with respect to the ILM lever.

Figure 3-4. Pin1 and ILM Lever



3.3 ILM Cover

As indicated in [Table A-4](#), ILM covers are available as discrete components and pre-assembled to the ILM load plate.

The ILM cover will interfere with a processor and pop off if the ILM is closed with a processor in the socket.

The ILM cover is designed to be interchangeable between different suppliers validated by Intel. Performance of the pop off feature may decline if the ILM cover supplier is different than the ILM supplier. The ILM cover can be removed manually if the pop off feature is not desirable, or not functional.

The ILM cover has UL94 V-0 flammability rating.

The ILM cover durability requirement is 20 cycles (1 cycle = install and remove).

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Independent Loading Mechanism (ILM) and Back Plate



4 LGA1356 Socket, ILM and Back Plate Electrical, Mechanical, and Environmental Specifications

This chapter describes the electrical, mechanical, and environmental specifications for the LGA1356 socket, Independent Loading Mechanism and Back Plate.

4.1 Component Mass

Table 4-1. Component Mass

Component	Mass
Socket Body, Contacts and PnP Cover	15 gm
ILM Assembly	43 gm
Back Plate	100 gm

4.2 Package / Socket Stackup Height

Table 4-2 provides the stackup height of a processor in the 1356-land LGA package and LGA1356 socket with the ILM closed and the processor fully seated in the socket.

Table 4-2. 1356-land Package and LGA1356 Socket Stackup Height

Integrated Stackup Height (mm) From Top of Board to Top of IHS	7.753 ± 0.262 mm
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Notes:

- This data is provided for information only, and is derived from: (a) the height of the socket seating plane above the motherboard after reflow, given in [Appendix C](#), (b) the height of the package, from the package seating plane to the top of the IHS, and accounting for its nominal variation and tolerances that are given in the corresponding processor EDS and expected values for the follow-on processor.
- This value is a RSS calculation.

4.3 Socket Maximum Temperature

The power dissipated within the socket is a function of the current at the pin level and the effective pin resistance. To ensure socket long term reliability, Intel defines socket maximum temperature using a via on the underside of the motherboard. Exceeding the temperature guidance may result in socket body deformation, or increases in thermal and electrical resistance which can cause a thermal runaway and eventual electrical failure. The guidance for socket maximum temperature is listed below:

- Via temperature under socket < 96 °C



4.4 Loading Specifications

The socket will be tested against the conditions listed in the LGA1366 Socket Validation Reports, and LGA1356 Addendum, with heatsink, ILM and back plate attached, under the loading conditions outlined in this chapter.

Table 4-3 provides load specifications for the LGA1356 socket with the ILM and back plate installed. The maximum limits should not be exceeded during heatsink assembly, shipping conditions, or standard use condition. Exceeding these limits during test may result in component failure. The socket body should not be used as a mechanical reference or load-bearing surface for thermal solutions.

Table 4-3. Socket and ILM Mechanical Specifications

Parameter	Min	Max	Notes
Static compressive load from ILM to processor IHS	445 N [100 lbf]	623 N [140 lbf]	3, 4
Thermal Solution Static Compressive Load	0 N [0 lbf]	266 N [60 lbf]	1, 2, 3
Total Static Compressive Load (ILM plus Heatsink)	445 N (100 lbf)	890 N (200 lbf)	3, 4
Dynamic Compressive Load (with heatsink installed)	N/A	890 N [200 lbf]	1, 3, 5, 6
Target Pick and Place Cover allowable removal force	N/A	4.45 - 6.68 N [1.0 - 1.5 lbf]	
Load Lever actuation force	N/A	38.3 N [8.6 lbf] in the vertical direction 10.2 N [2.3 lbf] in the lateral direction.	

Notes:

1. These specifications apply to uniform compressive loading in a direction perpendicular to the IHS top surface.
2. This is the minimum and maximum static force that can be applied by the heatsink and it's retention solution to maintain the heatsink to IHS interface. This does not imply the Intel reference TIM is validated to these limits. TIM load range is documented in [Section 5.2](#) for the Intel Reference Design.
3. Loading limits are for the LGA1356 socket.
4. This minimum limit defines the compressive force required to electrically seat the processor onto the socket contacts.
5. Dynamic loading is defined as an 11 ms duration average load superimposed on the static load requirement.
6. Test condition used a heatsink mass of 550 gm [1.21 lb] with 50 g acceleration measured at heatsink mass. The dynamic portion of this specification in the product application can have flexibility in specific values, but the ultimate product of mass times acceleration should not exceed this dynamic load.

4.5 Electrical Requirements

LGA1356 socket electrical requirements are measured from the socket-seating plane of the processor to the component side of the socket PCB to which it is attached. All specifications are maximum values (unless otherwise stated) for a single socket contact, but includes effects of adjacent contacts where indicated.

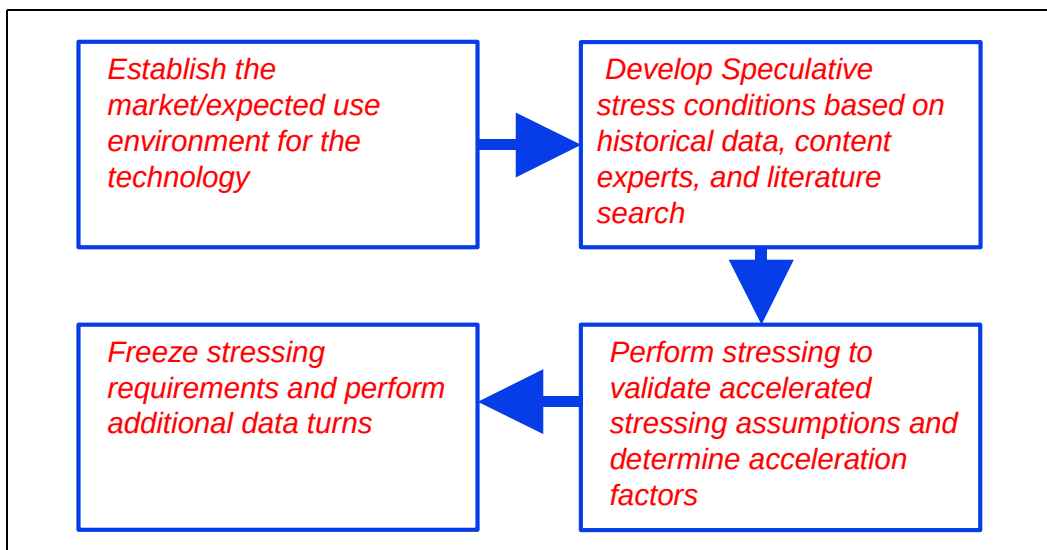
**Table 4-4. Electrical Requirements for LGA1356 Socket**

Parameter	Value	Comment
Mated loop inductance, Loop	<3.9 nH	The inductance calculated for two contacts, considering one forward conductor and one return conductor. These values must be satisfied at the worst-case height of the socket.
Maximum mutual capacitance, C.	<1 pF	The capacitance between two contacts
Socket Average Contact Resistance (EOL)	15.2 mΩ	The socket average contact resistance target is derived from average of every chain contact resistance for each part used in testing, with a chain contact resistance defined as the resistance of each chain minus resistance of shorting bars divided by number of lands in the daisy chain. The specification listed is at room temperature and has to be satisfied at all time. Socket Contact Resistance: The resistance of the socket contact, solderball, and interface resistance to the interposer land.
Max Individual Contact Resistance (EOL)	≤ 100 mΩ	The specification listed is at room temperature and has to be satisfied at all time. Socket Contact Resistance: The resistance of the socket contact, solderball, and interface resistance to the interposer land; gaps included.
Bulk Resistance Increase	≤ 3 mΩ	The bulk resistance increase per contact from 24 °C to 107 °C
Dielectric Withstand Voltage	360 Volts RMS	
Insulation Resistance	800 MΩ	

4.6 Environmental Requirements

The reliability targets in this chapter are based on the expected field use environment for these products. The test sequence for the LGA1356 socket was developed using the knowledge-based reliability evaluation methodology, which is acceleration factor dependent. A simplified process flow of this methodology can be seen in [Figure 4-1](#). Since the LGA1356 socket is very similar to the LGA1366 socket, the LGA1356 socket is expected to perform similarly and full validation for the LGA1356 socket is avoided.

Figure 4-1. Flow Chart of Knowledge-Based Reliability Evaluation Methodology



A detailed description of this methodology can be found at:

<ftp://download.intel.com/technology/itj/q32000/pdf/reliability.pdf>.

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5 Thermal Solutions

This section describes a 1U reference heatsink and thermal design guidelines for the Intel® Xeon® Processor E5-2400 Product Family.

5.1 Boundary Conditions

Table 5-1 provides values for boundary conditions and performance targets used to generate processor thermal specifications and to provide guidance for heatsink design.

Table 5-1. Values Used to Generate Processor Thermal Specifications

Parameter	Value					
Altitude, system ambient temp	Sea level, 35°C					
TDP	50W (4-core)	60W	70W	80W (4-core)	95W	80W (2-core, 1 socket)
Ψ_{CA}^1	0.312°C/W	0.296°C/W	0.296°C/W	0.315°C/W	0.296°C/W (8-core), 0.298°C/W (6-core)	0.285°C/W
T_{LA}^2	49°C					48.1°C
Airflow ³	9.7 CFM @ 0.23" dP					13 CFM @ 0.28" dP
System height (form factor)	1U (EEB) ⁴					1U (non-specific, 1-socket)
Heatsink volumetric ⁵	90 x 90 x 25.5 mm (1U/SSI blade) ⁶					
Heatsink technology ⁷	Cu base, Al fins					

Notes:

1. Max target (mean + 3 sigma + offset) for thermal characterization parameter (Section 5.4.1).
2. Local ambient temperature of the air entering the heatsink.
3. Airflow through the heatsink fins with zero bypass. Max target for pressure drop (dP) measured in inches H₂O.
4. Reference system configuration. Processor is downstream from memory in EEB (Entry-Level Electronics Bay). Values above do not apply to LR-DIMM in an Intel Reference Design. Ducting is utilized to direct airflow.
5. Dimensions of heatsink do not include socket or processor.
6. Heatsink height + socket/processor height (Table 4-2) complies with TEB 1U Rack Height Constraints (36 mm) in EEB Specification 2011, and with Maximum Component Height (33.5 mm) in SSI Compute Blade Specification, both at <http://www.ssiforum.org>.
7. Passive heatsinks. PCM45F thermal interface material.

Table 5-2 provides approximate boundary conditions and approximate performance expectations in Compact Electronics Bay. These values are not used to generate processor thermal specifications, but may provide guidance for heatsink design.

Table 5-2. Performance Expectations in Compact Electronics Bay (CEB)

Parameter	Value				
Altitude, system ambient temp	Sea level, 35°C				
TDP	50W	60W	70W	80W (4-core)	95W
T_{LA}^1	43.7°C	45.6°C	46.8°C	48.1°C	50.0°C (8-core), 46.6°C (6-core)

Table 5-2. Performance Expectations in Compact Electronics Bay (CEB)

Parameter	Value				
Ψ_{CA}^2	0.273°C/W	0.265°C/W	0.264°C/W	0.278°C/W	0.265°C/W (8-core), 0.269°C/W (6-core)
Airflow ³	13 CFM @ 0.32" dP				
System height (form factor) ⁴	1U (CEB)				
Heatsink volumetric ⁵	90 x 90 x 25.5 mm (1U/SSI blade) ⁶				
Heatsink technology ⁷	Cu base, Al fins				

Notes:

1. Local ambient temperature of the air entering the heatsink.
2. Max target (mean + 3 sigma + offset) for thermal characterization parameter (Section 5.4.1).
3. Airflow through the heatsink fins with zero bypass. Max target for pressure drop (dP) measured in inches H₂O.
4. Reference system configuration. Processor is downstream from processor in CEB (Compact Electronics Bay). With the values above, the 25.5mm tall heatsink can meet the processor thermal specifications in Intel's Reference Design 10.5x12 inches CEB board. However, these CEB values are not used to generate processor thermal specifications. Ducting is utilized to direct airflow.
5. Dimensions of heatsink do not include socket or processor.
6. Heatsink height + socket/processor height (Table 4-2) complies with TEB 1U Rack Height Constraints (36 mm) in EEB Specification 2011, and with Maximum Component Height (33.5 mm) in SSI Compute Blade Specification, both at <http://www.ssi-forum.org>.
7. Passive heatsinks. PCM45F thermal interface material.

Table 5-1 and Table 5-2 specify Ψ_{CA} and pressure drop targets for specific airflows. To determine Ψ_{CA} and pressure drop targets for other airflows, use Best-fit equations in Figure 5-1. Heatsink detailed drawings are in Appendix A.

Figure 5-1. Best-fit Equations

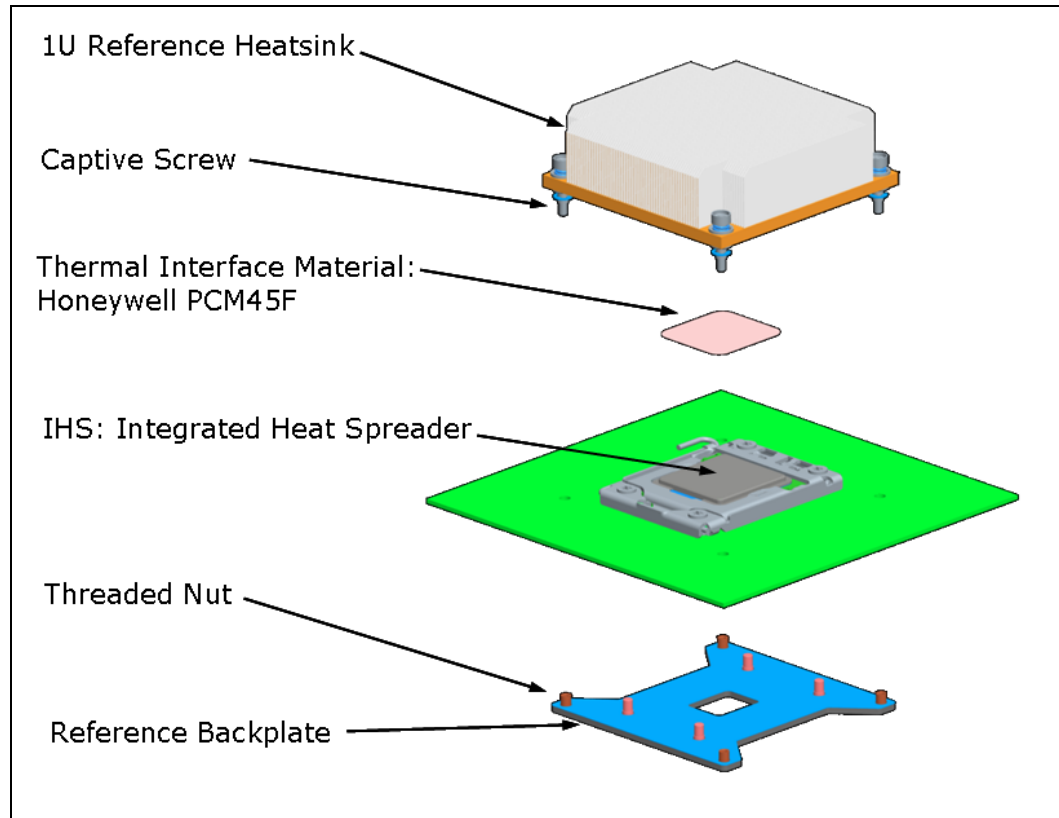
	Sigma	α	β	γ	a_quad	b_linear
25.5mm	0.0057	0.176	1.65	1.08	2.41E-04	2.15E-02
1U	0.0046	0.168	1.61	1.04	1.30E-04	1.97E-02
2U	0.0059	0.132	1.39	0.986	7.2E-05	3.6E-03
Tower	0.0055	0.131	1.515	0.995	3.74E-5	4.69E-3

$$\Psi_{ca} \text{ (mean)} = \alpha + \beta * (\text{CFM})^{-\gamma}$$

$$\Delta P = a_quad * (\text{CFM})^2 + b_linear * (\text{CFM})$$

5.2 Assembly

Figure 5-2. 1U Reference Heatsink Assembly



The assembly process for the 1U reference heatsink begins with application of Honeywell PCM45F thermal interface material to improve conduction from the IHS. Tape and roll format is recommended. Pad size is 35 x 35 mm, thickness is 0.25 mm.

Next, position the heatsink such that the heatsink fins are parallel to system airflow. While lowering the heatsink onto the IHS, align the four captive screws of the heatsink to the four threaded nuts of the back plate.

Using a #2 Phillips driver, torque the four captive screws to 8 inch-pounds. Fastener sequencing, in other words starting the threads on all four screws before torquing, may mitigate against cross threading.

This assembly process is designed to produce a static load of 39 - 51 lbf, for 0.062" - 0.100" board thickness respectively. Honeywell PCM45F is expected to meet the performance targets in [Table 5-1](#) and [Table 5-2](#) from 30 - 60 lbf. From [Table 4-3](#), the Heatsink Static Compressive Load of 0 - 60 lbf allows for designs that vary from the 1U reference heatsink. Example: A customer's unique heatsink with very little static load (as little as 0 lbf) is acceptable from a socket loading perspective as long as the thermal specifications are met.

Compliance to Board Keepout Zones in [Appendix A](#) is assumed for this assembly process.

5.2.1 Thermal Interface Material (TIM)

TIM should be verified to be within its recommended shelf life before use.

Surfaces should be free of foreign materials prior to application of TIM.

Use isopropyl alcohol and a lint free cloth to remove old TIM before applying new TIM.

5.3 Structural Considerations

Target mass of heatsinks should not exceed 500 gm.

From [Table 4-3](#), the Dynamic Compressive Load of 200 lbf max allows for designs that exceed 500 gm as long as the mathematical product does not exceed 200 lbf. Example: A heatsink of 2-lb mass (908 gm) x 50 g (acceleration) x 2.0 Dynamic Amplification Factor = 200 lbf. The Total Static Compressive Load ([Table 4-3](#)) should also be considered in dynamic assessments.

Direct contact between back plate and chassis pan will help minimize board deflection during shock. Placement of board-to-chassis mounting holes also impacts board deflection and resultant socket solder ball stress. Customers need to assess shock for their designs as their heatsink retention (back plate), heatsink mass and chassis mounting holes may vary.

5.4 Thermal Design

5.4.1 Thermal Characterization Parameter

The case-to-local ambient Thermal Characterization Parameter (Ψ_{CA}) is defined by:

Equation 5-1. $\Psi_{CA} = (T_{CASE} - T_{LA}) / TDP$

Where:

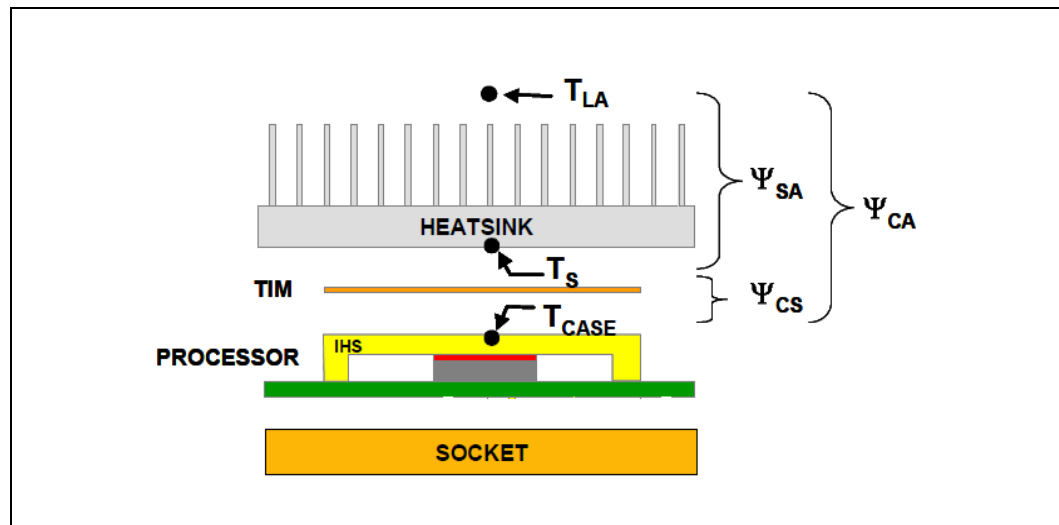
- T_{CASE} = Processor case temperature (°C). For T_{CASE} specification see the appropriate External Design Specification (EDS).
- T_{LA} = Local ambient temperature in chassis at processor (°C).
- TDP = TDP (W) assumes all power dissipates through the integrated heat spreader. This inexact assumption is convenient for heatsink design. TTVs are often used to dissipate TDP. Correction offsets account for differences in temperature distribution between processor and TTV.

Equation 5-2. $\Psi_{CA} = \Psi_{CS} + \Psi_{SA}$

Where:

- Ψ_{CS} = Thermal characterization parameter of the TIM (°C/W) is dependent on the thermal conductivity and thickness of the TIM.
- Ψ_{SA} = Thermal characterization parameter from heatsink-to-local ambient (°C/W) is dependent on the thermal conductivity and geometry of the heatsink and dependent on the air velocity through the heatsink fins.

[Figure 5-3](#) illustrates the thermal characterization parameters.

Figure 5-3. Processor Thermal Characterization Parameter Relationships


5.5 Fan Speed Control

5.5.1 Fundamentals

In server platforms, processors often share airflow provided by system fans with other system components such as chipset, memory and hard drives. As such, the thermal control features in chipset, memory and other components not covered in this document, should influence system fan speed control to reduce fan power consumption and help systems meet acoustic targets.

The addition of thermal sensors placed in the system (for example, on front panel or motherboard) to augment internal device sensors (for example, in processor, chipset and memory) will improve the ability to implement need-based fan speed control. The placement of system sensors in cooling zones, where each zone has dedicated fan(s), can improve the ability to tune fan speed control for optimal performance and/or acoustics.

System events such as fan or power supply failure, device events such as TCC Activation or THERMTRIP, and maintenance events such as hot swap time allowance, need to be comprehended to implement appropriate fan speed control to prevent undesirable performance or loss of data. For more information on device events and features see the appropriate processor Datasheet.

Tcontrol and its upper and lower limits defined by hysteresis, can be used to avoid fan speed oscillation and undesirable noise variations.

5.6 Thermal Features

More information regarding processor thermal features is contained in the appropriate datasheet.

5.6.1 T_{CONTROL} and DTS Relationship

Improved acoustics and lower fan power can be achieved by understanding the T_{CONTROL} and DTS relationship, and implementing fan speed control accordingly.

Table 5-3. T_{CONTROL} and DTS Relationship

Condition	Fan Speed Control
$DTS \leq T_{\text{CONTROL}}$	Adjust fan speed to maintain $DTS \leq T_{\text{CONTROL}}$.
$DTS > T_{\text{CONTROL}}$	Adjust fan speed to keep T_{CASE} at or below the T_{CASE} based thermal profile in the EDS, or adjust fan speed to keep DTS at or below the DTS based thermal profile in the EDS.

5.6.1.1 Sign Convention and Temperature Filtering

Digital Thermal Sensor (DTS) and T_{CONTROL} are relative die temperatures offset below the Thermal Control Circuit (TCC) activation temperature. As such, negative sign conventions are understood. While DTS and T_{CONTROL} are available over PECI and MSR, use of these values in fan speed control algorithms requires close attention to sign convention. See Table 5-4 for the sign convention of various sources.

Table 5-4. Sign Convention

	MSR (BWG)	PECI (EDS)
DTS	(+) using PACKAGE_THERM_STATUS (22:16, Digital Readout)	(-) using GetTemp()
T_{CONTROL}	(+) using TEMPERATURE_TARGET (15:8, Temperature Control Offset)	(+) using Temperature Target Read from RdPkgConfig()

Where a positive (+) sign convention is shown in Table 5-4, no sign bit is actually assigned, so writers of firmware code may mistakenly assign a positive sign convention in firmware equations. As appropriate, a negative sign should be introduced.

Where a negative (-) sign convention is shown in Table 5-4, a sign bit is assigned, so firmware code will read a negative sign convention in firmware equations, as desired.

DTS obtained thru MSR (PACKAGE_THERM_STATUS) is an instantaneous value. As such, temperature readings over short time intervals may vary considerably using this MSR. For this reason, DTS obtained thru PECI GetTemp() may be preferred since temperature filtering will provide the thermal trend.

5.6.1.2 T_{CONTROL} Relief

Factory configured T_{CONTROL} values are available in the appropriate Dear Customer Letter or may be extracted by issuing a Mailbox or an RDMSR instruction. See the appropriate External Design Specification (EDS) for more information.

Due to increased thermal headroom based on thermal characterization on the latest processors, customers have the option to reduce T_{CONTROL} to values lower than the factory configured values.

In some situations, use of T_{CONTROL} Relief can reduce average fan power and improve acoustics. There are no plans to change Intel's specification or the factory configured T_{CONTROL} values on individual processors.



To implement this relief, customers must re-write code to set T_{CONTROL} to the reduced values provided in the table below. Implementation is optional. Alternately, the factory configured T_{CONTROL} values can still be used, or some value between factory configured and Relief. Regardless of T_{CONTROL} values used, BIOS needs to identify the processor type.

Table 5-5. T_{CONTROL} Relief for Intel® Xeon® Processor E5-2400 Product Family

TDP, # Core	T_{CONTROL} Relief	Max Core Frequency	Factory Configured
95W 8C	-6	2.30 GHz or lower	-10
95W 6C	-6	2.40 GHz or lower	-10
70W 8C	-6	1.80 GHz or lower	-10
60W 6C	-6	2.00 GHz or lower	-10
80W 4C	-6	2.20 GHz or lower	-10
80W 2C, 1S	-6	2.80 GHz or lower	-10

In some cases, use of Tcontrol Relief as the trigger point for fan speed control may result in excessive TCC activation. To avoid this, the adjusted trigger point for fan speed control (FSC) is defined as:

$$T_{\text{control_FSC}} = -T_{\text{CONTROL}} + T_{\text{control_offset}}$$

$T_{\text{control_offset}}$ must be chosen such that $T_{\text{control_FSC}} < T_{\text{control Relief}}$. As such, $T_{\text{control_FSC}}$ is an earlier trigger point for fan speed control, as compared to $T_{\text{control Relief}}$, and can be interpreted as overcooling. When overcooling to $T_{\text{control_FSC}}$, margin as defined in [Section 5.8.3](#) and [Section 5.8.6](#) can be ignored. As compared to cooling to $T_{\text{control Relief}}$, overcooling to $T_{\text{control_FSC}}$:

- May increase frequency benefit from Intel TBT as defined in [Section 5.6.3](#).
- Will increase acoustics
- May result in lower wall power

Customers must characterize a $T_{\text{control_offset}}$ value for their system to meet their goals for frequency, acoustics and wall power.

5.6.2 Short Duration TCC Activation and Catastrophic Thermal Management for Intel® Xeon® Processor E5-2400 Product Family

Systems designed to meet thermal capacity may encounter short durations of throttling, also known as TCC activation, especially when running non-steady processor stress applications. This is acceptable and is functionally within the intended temperature control parameters of the processor. Such short duration TCC activation is not expected to provide noticeable reductions in application performance, and is typically within the normal range of processor to processor performance variation. Normal amounts of TCC activation occur at PECI values less than -0.25. Such occurrences may cause utilities or operating systems to issue error log.

PECI = -0.25 indicates a catastrophic thermal failure condition in all studies conducted. As such, to help prevent loss of data, a soft shutdown can be initiated at PECI = -0.25. Since customer designs, boundary conditions, and failure scenarios differ, this guidance should be tested in the customer's system to prevent loss of data during shutdown. PECI command GetTemp() can be used to obtain non-integer PECI values.

5.6.3 Intel® Turbo Boost Technology

Intel® Turbo Boost Technology (Intel® TBT), available on certain processor SKUs, opportunistically, and automatically, allows the processor to run faster than the marked frequency if the part is operating below its power, temperature and current limits.

Heatsink performance (lower Ψ_{CA} as described in [Section 5.4.1](#)) is one of several factors that can impact the amount of Intel TBT frequency benefit. Intel TBT performance is also constrained by ICC, and VCC limits.

Increased IMON accuracy may provide more Intel TBT benefit on TDP limited applications, as compared to lower Ψ_{CA} , as temperature is not typically the limiter for these workloads.

With Intel TBT enabled, the processor may run more consistently at higher power levels (but still within TDP), and be more likely to operate above $T_{CONTROL}$, as compared to when Intel TBT is disabled. This may result in higher acoustics.

5.7 Thermal Guidance

5.7.1 Thermal Excursion

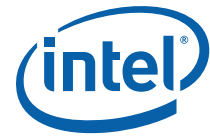
Under fan failure or other anomalous thermal excursions, T_{case} may exceed the thermal profile for a duration totaling less than 360 hours per year without affecting long term reliability (life) of the processor. For more typical thermal excursions, Thermal Monitor is expected to control the processor power level as long as conditions do not allow the T_{case} to exceed the temperature at which Thermal Control Circuit (TCC) activation initially occurred. Under more severe anomalous thermal excursions when the processor temperature cannot be controlled at or below this T_{case} level by TCC activation, then data integrity is not assured. At some higher threshold, THERMTRIP_N will enable a shut down in an attempt to prevent permanent damage to the processor. Thermal Test Vehicle (TTV) may be used to check anomalous thermal excursion compliance by ensuring that the processor T_{case} value, as measured on the TTV, does not exceed T_{case_max} at the anomalous power level for the environmental condition of interest. This anomalous power level is equal to 75% of the Thermal Design Power (TDP) limit.

This guidance can be applied to 95W, 80W, 70W, 60W Standard or Basic SKUs in the Intel® Xeon® Processor E5-2400 Product Family.

5.7.2 Absolute Processor Temperature

Intel does not test any third party software that reports absolute processor temperature. As such, Intel cannot recommend the use of software that claims this capability. Since there is part-to-part variation in the TCC (thermal control circuit) activation temperature, use of software that reports absolute temperature can be misleading.

See the appropriate Datasheet for details regarding use of TEMPERATURE_TARGET register to determine the minimum absolute temperature at which the TCC will be activated and PROCHOT# will be asserted.



5.8 DTS Based Thermal Specification

5.8.1 Compliance to Tcase Based Thermal Profile

Processor heatsink design must still comply with the Tcase based thermal profile provided in the *Intel® Xeon® Processor E5-2400 Product Family Datasheet - Volume One*. Heatsink design compliance can be determined with thermocouple and TTV as with previous processors.

The heat sink is sized to comply with the Tcase based thermal profile. Customers have an option to either follow processor based Tcase spec or follow the DTS based thermal specification. In some situations, implementation of DTS based thermal specification can reduce average fan power and improve acoustics as compared to the Tcase based thermal profile.

When all cores are active, a properly sized heatsink will be able to meet the DTS based thermal specification. When all cores are not active or when Intel Turbo Boost Technology is active, attempting to comply with the DTS based thermal specification may drive system fans to maximum speed. In such situations, the T_{CASE} temperature will be below the T_{CASE} based thermal profile by design.

5.8.2 Considerations for Follow-on Processor

The follow-on processor in the platform will have new capabilities as compared to the Intel® Xeon® Processor E5-2400 Product Family. For example, the follow-on processor has a new Package Configuration Space (PCS) command to read margin (M) from the processor: `RdPkgConfig()`, Index 10. For the Intel® Xeon® Processor E5-2400 Product Family, margin (M) must be calculated in firmware.

In the following sections, implementation details specified for the Intel® Xeon® Processor E5-2400 Product Family can also be used for the follow-on processor.

For more information regarding the differences between the follow-on processor and the Intel® Xeon® Processor E5-2400 Product Family see *Platform Digital Thermal Sensor (DTS) Based Thermal Specifications and Overview*.

5.8.3 DTS Based Thermal Profile, Tcontrol and Margin for the Intel® Xeon® Processor E5-2400 Product Family

The calculation of the DTS based thermal specification is based on both Tcontrol and the DTS Based Thermal Profile (T_{DTS}):

$$T_{DTS} = \min[T_{LA} + \Psi_{pa} * P * F, \text{TEMPERATURE_TARGET [23:16]} - Tcc_Offset]$$

Where $T_{LA} + \Psi_{pa}$ are the intercept and slope terms from the T_{DTS} equations in the appropriate External Design Specification (EDS). To implement the DTS based thermal specification, these equations must be programmed in firmware. Since the equations differ with processor SKU, SKUs can be identified by TDP, Core Count and a profile identifier (CSR bits). For associated commands, see *Platform Digital Thermal Sensor (DTS) Based Thermal Specifications and Overview*.

Power (P) is calculated in [Section 5.8.4](#). As power dynamically changes, the specification also changes, so power and T_{DTS} calculations are recommended every 1 second.

Correction factor (F) compensates for the error in power monitoring. The current estimate for F is 0.95.

The Tcontrol portion of the DTS based thermal specification is a one time calculation:

$$T_{\text{control_spec}} = \text{TEMPERATURE_TARGET} [23:16] - T_{\text{control}} + T_{\text{control_offset}}$$

Tcontrol is defined in [Section 5.6.1.1](#). Tcontrol_offset is defined in [Section 5.6.1.2](#).

The final DTS based thermal specification is the maximum of both:

$$T_{\text{DTS_max}} = \max[T_{\text{control_spec}}, T_{\text{DTS}}]$$

The margin (M) between the actual die temperature and the DTS based thermal specification is used in the fan speed control algorithm. When $M < 0$, increase fan speed. When $M \geq 0$, fan speed may decrease.

$$M = T_{\text{DTS_max}} - T_{\text{sensor}}$$

OR

$$M = T_{\text{DTS_ave}} - T_{\text{sensor}}$$

Tsensor represents the absolute temperature of the processor as power changes:

$$T_{\text{sensor}} = \text{TEMPERATURE_TARGET} [23:16] + \text{DTS}$$

T_{DTS_ave} is defined in [Section 5.8.5](#).

TEMPERATURE_TARGET [23:16], the temperature at which the processor thermal control circuit activates, is a one time PECL readout: RdPkgConfig(), Temperature Target Read, 23:16.

DTS, the relative temperature from thermal control circuit activation, is negative by definition, and changes instantaneously. DTS command info is given in [Section 5.6.1.1](#).

5.8.4 Power Calculation for the Intel® Xeon® Processor E5-2400 Product Family

To implement DTS based thermal specification, average power over time must be calculated:

$$P = (E2 - E1) / (t2 - t1)$$

Where:

t1 = time stamp 1

t2 = time stamp 2

E1 = Energy readout at time t1

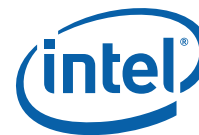
E2 = Energy readout at time t2

The recommended time interval between energy readings is 1 second. This helps ensure the power calculation is accurate by making the error between time stamps small as compared to the duration between time stamps.

For details regarding energy readings, see *Platform Digital Thermal Sensor (DTS) Based Thermal Specifications and Overview*.

5.8.5 Averaging the DTS Based Thermal Specification for the Intel® Xeon® Processor E5-2400 Product Family

Averaging the DTS Based Thermal Specification helps keep the rate of change of the temperature specification on the same scale as the actual processor temperature, and helps avoid rapid changes in fan speed when power changes rapidly.



An exponential average of the specification can be calculated using a two time constant model:

$$T_{DTS_f} = \alpha_f \times DT \times T_{DTS_max} + T_{DTS_f_previous} \times (1 - \alpha_f \times DT)$$

$$T_{DTS_s} = \alpha_s \times DT \times T_{DTS_max} + T_{DTS_s_previous} \times (1 - \alpha_s \times DT)$$

$$T_{DTS_ave} = C \times T_{DTS_f} + (1 - C) \times T_{DTS_s}$$

Where:

T_{DTS_max} is the instantaneous spec
 T_{DTS_f} and T_{DTS_s} are the fast and slow time averages
 T_{DTS_ave} is the final two time constant average specification
 α_f and α_s are the time constant coefficients
 C is a scale factor
 DT is the scan rate and is recommended to be approximately 1 second

Table 5-6 below shows the coefficients recommended for averaging. These values may change per processor SKU. Customers should tune these coefficients based on their thermal solutions.

Table 5-6. Averaging Coefficients

Heatsink Performance	α_f (1/s)	α_s (1/s)	C	Comment
Low	1.0	0.04	0.30	based on typical processor
Medium	1.0	0.07	0.30	based on typical processor
High	1.0	0.10	0.40	based on typical processor

5.8.6 Capabilities for the Follow-on Processor

For the follow-on processor, the intercept and slope terms from the T_{DTS} equations (T_{LA} , Ψ_{pa}), as defined in Section 5.8.3, are stored in the processor. This allows margin (M) to be reported by the processor. The PECL command for margin (M) will be RdPkgConfig(), Index 10.

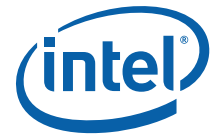
$M < 0$; gap to spec, fan speed must increase
 $M \geq 0$; margin to spec, fan speed may decrease

Use of RdPkgConfig(), Index 10 with the Intel® Xeon® Processor E5-2400 Product Family will return an illegal command.

For the follow-on processor, coefficients (α_f , α_s), scale factor (C) and correction factor (F) will be factory configured.

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6 Quality and Reliability Requirements

6.1 Test Conditions

Test Conditions, Qualification and Visual Criteria vary by customer.

Socket Test Conditions are provided in the LGA1366 Socket Validation Reports, and LGA1356 Addendum and are available from socket suppliers listed in [Appendix A](#).

6.2 Intel Reference Component Validation

Intel tests reference components both individually and as an assembly on mechanical test boards, and assesses performance to the envelopes specified in previous sections by varying boundary conditions.

While component validation shows that a reference design is tenable for a limited range of conditions, customers need to assess their specific boundary conditions and perform reliability testing based on their use conditions.

Intel reference components are also used in board functional tests to assess performance for specific conditions.

6.2.1 Board Functional Test Sequence

Each test sequence should start with components (baseboard, heatsink assembly, and so on) that have not been previously submitted to any reliability testing.

The test sequence should always start with a visual inspection after assembly and BIOS/Processor/memory test. The stress test should be then followed by a visual inspection and then BIOS/Processor/memory test.

6.2.2 Post-Test Pass Criteria

The post-test pass criteria are:

1. No significant physical damage to the heatsink and retention hardware.
2. Heatsink remains seated and its bottom remains mated flat against the IHS surface. No visible gap between the heatsink base and processor IHS. No visible tilt of the heatsink with respect to the retention hardware.
3. No signs of physical damage on baseboard surface due to impact of heatsink.
4. No visible physical damage to the processor package.
5. Successful BIOS/Processor/memory test.
6. Thermal compliance testing to demonstrate that the case temperature specification can be met.



6.2.3 Recommended BIOS/Processor/Memory Test Procedures

This test is to ensure proper operation of the product before and after environmental stresses, with the thermal mechanical enabling components assembled. The test shall be conducted on a fully operational baseboard that has not been exposed to any battery of tests prior to the test being considered.

The testing setup should include the following components, properly assembled and/or connected:

- Appropriate system baseboard.
- Processor and memory.
- All enabling components, including socket and thermal solution parts.

The pass criterion is that the system under test shall successfully complete the checking of BIOS, basic processor functions and memory, without any errors.

6.3 Material and Recycling Requirements

Material shall be resistant to fungal growth. Examples of non-resistant materials include cellulose materials, animal and vegetable based adhesives, grease, oils, and many hydrocarbons. Synthetic materials such as PVC formulations, certain polyurethane compositions (for example, polyester and some polyethers), plastics which contain organic fillers of laminating materials, paints, and varnishes also are susceptible to fungal growth. If materials are not fungal growth resistant, then MIL-STD-810E, Method 508.4 must be performed to determine material performance.

Any plastic component exceeding 25 gm should be recyclable per the European Blue Angel recycling standards.

The following definitions apply to the use of the terms lead-free, Pb-free, and RoHS compliant.

Lead-free and Pb-free: Lead has not been intentionally added, but lead may still exist as an impurity below 1000 ppm.

RoHS compliant: Lead and other materials banned in RoHS Directive are either (1) below all applicable substance thresholds as proposed by the EU or (2) an approved/pending exemption applies.

Note: RoHS implementation details are not fully defined and may change.





A Component Suppliers

Various suppliers have developed support components for processors in the Intel® Xeon® Processor E5-2400 Product Family-based platform. These suppliers and components are listed as a convenience to customers. Intel does not guarantee quality, reliability, functionality or compatibility of these components. The supplier list and/or the components may be subject to change without notice. Customers are responsible for the thermal, mechanical, and environmental verification of the components with the supplier.

A.1 Intel Enabled Supplier Information

Performance targets for heatsinks are described in [Section 5.1](#). Mechanical drawings are provided in [Appendix A](#). Mechanical models are listed in [Table 1-1](#). Heatsinks assemble to server back plate [Table A-4](#).

A.1.1 Intel Reference Thermal Solution

Customers can purchase the Intel reference thermal solutions from the suppliers listed in [Table A-1](#).

Table A-1. Suppliers for the Intel Reference Thermal Solution

Assembly	Component	Description	Supplier PN	Supplier Contact Info
Assembly, Heat Sink, Intel Xeon processor E5-2400 product family, 1U	1U URS Intel Reference Heatsink p/n E32409-001	27 mm 1U Aluminum Fin, Copper Base, includes TIM, capable up to 95W	Fujikura HSA-8078 Rev A	Fujikura America Yuji Yasuda yuji@fujikura.com 408-748-6991
	1U URS SSI Blade Reference Heatsink p/n E39069-001 refers to E22056 Rev 02 + Snap Cover	25.5 mm 1U Aluminum Fin, Copper Base, includes TIM and Snap Cover, capable up to 95W	Fujikura HSA-8083C	Fujikura Taiwan Branch Yao-Hsien Huang yeohsien@fujikuratw.com.tw 886(2)8788-4959
		Thermal Interface Material	Honeywell PCM45F	Honeywell International, Inc. Judy Oles (Customer Service) Judy.Oles@Honeywell.com 509-252-8605 Andrew S.K. Ho (APAC) andrew.ho@honeywell.com (852) 9095-4593 Andy Delano (Technical) Andrew.Delano@Honeywell.com 509-252-2224

A.1.2 Intel Collaboration Thermal Solution

Customers can purchase the Intel collaboration thermal solutions from the suppliers listed in [Table A-2](#).



Table A-2. Suppliers for the Intel Collaboration Thermal Solution

Assembly	Component	Description	Supplier PN	Supplier Contact Info
Assembly, Heatsink, Intel Xeon processor E5-2400 product family, 2U	2U URS Heatsink Intel Collaboration Heatsink p/n E32410-001	Supplier Designed Solution with Intel-specified retention, includes TIM, up to 95W capable	Foxconn pn 1A016500	Foxconn Ray Wang ray.wang@foxconn.com (512) 670-2638 ext 273
Assembly, Heatsink, Intel Xeon processor E5-2400 product family, Pedestal	Tower URS Heatsink Intel Collaboration Heatsink p/n E32412-001	Supplier Designed Solution with Intel-specified retention, includes TIM, up to 95W capable	Chaun-Choung Technology Corp (CCI) pn 0007029401	Chaun-Choung Technology Corp (CCI) Monica Chih monica_chih@ccic.com.tw +886 (2) 2995-2666 x1131 Sean Wu sean_wu@ccic.com.tw 408-768-7629

A.1.3 Alternative Thermal Solution

Customers can purchase the alternative thermal solutions from the suppliers listed in Table A-3.

Table A-3. Suppliers for the Alternative Thermal Solution (Sheet 1 of 3)

Assembly	Component	Description	Supplier PN	Thermal Capability
Assembly, Heat Sink, 1U	1U SSI Blade (25.5mm) Alternative URS Heatsink	Standard	TaiSol Corporation 1A1-9031000960-A www.Taisol.com	not capable for 80W (2-core, 1 socket); capable for all other SKUs up to 95W
		Standard	Thermaltake CL-P0484 www.Thermaltake.com	not capable for 80W (2-core, 1 socket); capable for all other SKUs up to 95W



Table A-3. Suppliers for the Alternative Thermal Solution (Sheet 2 of 3)

Assembly	Component	Description	Supplier PN	Thermal Capability
Assembly Heatsink, 1U	1U (27mm) Alternative URS Heatsink	Standard	CoolerMaster S1N-PJFCS-07-GP www.CoolerMaster.com	up to 95W capable
		Standard	Aavid Thermalloy 050073 www.AavidThermalloy.com	up to 95W capable
		Performance	Aavid Thermalloy 050231 www.AavidThermalloy.com	up to 95W capable
		Performance	Aavid Thermalloy 050232 www.AavidThermalloy.com	up to 95W capable
		Standard	CoolJag JYC0B39CTA www.CoolJag.com	up to 95W capable
		Performance	Taiwan Microloops 99-520040-M03 www.Microloops.com	up to 95W capable
		Performance	Asia Vital Components SQ42H00001 www.avc.com.tw	up to 95W capable
		Performance	Dynatron G218 www.Dynatron-Corp.com	up to 95W capable
		Performance	Delta Electronics DHS-B9090-20 www.deltaww.com	up to 95W capable
		Performance	Celsia Technologies 01IN001 www.celsiatechnologies.com	up to 95W capable
Assembly, Heatsink, 2U	2U Alternative URS Heatsink	Standard	Asia Vital Components SR40400001 www.avc.com.tw	up to 95W capable
		Standard	Asia Vital Components SR41400002 www.avc.com.tw	up to 95W capable
		Standard	Thermaltake CL-P0486 www.Thermaltake.com	up to 95W capable
		Standard	CoolerMaster S2N-PJMHS-07-GP www.CoolerMaster.com	up to 95W capable
		Standard	TaiSol Corporation 1A0-9041000960-A www.TaiSol.com	up to 95W capable
		Low Cost	Dynatron Corporation (Top Motor/Dynaeon) G520 www.Dynatron-Corp.com	not capable for 80W (2-core, 1 socket); capable for all other SKUs up to 95W
		Low Cost	CoolJag JAC0B40A www.CoolJag.com	not capable for 80W (2-core, 1 socket); capable for all other SKUs up to 95W



Table A-3. Suppliers for the Alternative Thermal Solution (Sheet 3 of 3)

Assembly	Component	Description	Supplier PN	Thermal Capability
Assembly, Heatsink, Tower	Tower Alternative URS Heatsink	Standard	TaiSol Corporation 1A0-9051000960-A www.Taisol.com	up to 95W capable
		Standard	Thermaltake CL-P0485 www.Thermaltake.com	up to 95W capable
		Standard	Asia Vital Components SS40W00001 www.avc.com.tw	up to 95W capable
Assembly, Heatsink	Pedestal/ 2U Active Heatsink	Active	Dynatron Corporation* (Top Motor/Dynaeon) G555 www.Dynatron-Corp.com	up to 95W capable

Notes:

- 1) Standard - Design and technology similar to Intel Reference or Collaboration designs, however, may not meet thermal requirements for all processor SKUs.
- 2) Performance - 1U Heatsink designed with premium materials or technology expected to provide optimum thermal performance for all processor SKUs.
- 3) Low Cost - 2U Cost-Optimized Heatsink, expected to meet thermal targets for lower power processor SKUs.

A.1.4 Socket, ILM and Back Plate

The LGA1356 Socket, ILM and Back Plate are described in [Chapter 2](#) and [Chapter 3](#), respectively. Socket mechanical drawings are provided in [Appendix C](#). Mechanical models are listed in [Table 1-1](#).

Table A-4. LGA1356 Socket, ILM and Back Plate

Item	Intel PN	Foxconn	Tyco	Molex
ILM Assembly	D92428-003	PT44L13-4102	1554105-1	475939000
ILM Assembly with ILM Cover	G13666-001	PT44L13-4111	1-1554105-1	475939070
ILM Cover	G14954-001	012-1000-5776	1-2134711-1	475930403
Back Plate	D92433-002	PT44P12-4104	1981467-2	475937000
LGA1356 Socket	E81085-001	PE135627-4371-01H	1554116-1	475943001
Supplier Contact Info		Julia Jiang juliaj@foxconn.com 408-919-6178	Billy Hsieh billy.hsieh@tycoelectronics.com +81 44 844 8292	Carol Liang carol.liang@molex.com Tel #: +86-21-5048-0889 ext 3301

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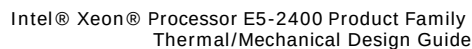


B Mechanical Drawings

Table B-1. Mechanical Drawing List

Description	Figure
Board Keepin / Keepout Zones (Sheet 1 of 4)	Figure B-1
Board Keepin / Keepout Zones (Sheet 2 of 4)	Figure B-2
Board Keepin / Keepout Zones (Sheet 3 of 4)	Figure B-3
Board Keepin / Keepout Zones (Sheet 4 of 4)	Figure B-4
1U Reference Heatsink Assembly (Sheet 1 of 2)	Figure B-5
1U Reference Heatsink Assembly (Sheet 2 of 2)	Figure B-6
1U Reference Heatsink Fin and Base (Sheet 1 of 2)	Figure B-7
1U Reference Heatsink Fin and Base (Sheet 2 of 2)	Figure B-8
Heatsink Shoulder Screw (1U, 2U and Tower)	Figure B-9
Heatsink Compression Spring (1U, 2U and Tower)	Figure B-10
Heatsink Retaining Ring (1U, 2U and Tower)	Figure B-11
Heatsink Load Cup (1U, 2U and Tower)	Figure B-12
2U Collaborative Heatsink Assembly (Sheet 1 of 2)	Figure B-13
2U Collaborative Heatsink Assembly (Sheet 2 of 2)	Figure B-14
2U Collaborative Heatsink Volumetric (Sheet 1 of 2)	Figure B-15
2U Collaborative Heatsink Volumetric (Sheet 2 of 2)	Figure B-16
Tower Collaborative Heatsink Assembly (Sheet 1 of 2)	Figure B-17
Tower Collaborative Heatsink Assembly (Sheet 2 of 2)	Figure B-18
Tower Collaborative Heatsink Volumetric (Sheet 1 of 2)	Figure B-19
Tower Collaborative Heatsink Volumetric (Sheet 2 of 2)	Figure B-20
1U Reference Heatsink Assembly with TIM (Sheet 1 of 2)	Figure B-21
1U Reference Heatsink Assembly with TIM (Sheet 2 of 2)	Figure B-22
2U Reference Heatsink Assembly with TIM (Sheet 1 of 2)	Figure B-23
2U Reference Heatsink Assembly with TIM (Sheet 2 of 2)	Figure B-24
Tower Reference Heatsink Assembly with TIM (Sheet 1 of 2)	Figure B-25
Tower Reference Heatsink Assembly with TIM (Sheet 2 of 2)	Figure B-26
25.5 mm Reference Heatsink Assembly (Sheet 1 of 2)	Figure B-27
25.5 mm Reference Heatsink Assembly (Sheet 2 of 2)	Figure B-28
25.5 mm Reference Heatsink Fin and Base (Sheet 1 of 2)	Figure B-29
25.5 mm Reference Heatsink Fin and Base (Sheet 2 of 2)	Figure B-30
25.5 mm Reference Heatsink Assembly with TIM (Sheet 1 of 2)	Figure B-31
25.5 mm Reference Heatsink Assembly with TIM (Sheet 2 of 2)	Figure B-32

52



Intel® Xeon® Processor E5-2400 Product Family
Thermal/Mechanical Design Guide

Figure B-3. Board Keepin / Keepout Zones (Sheet 3 of 4)

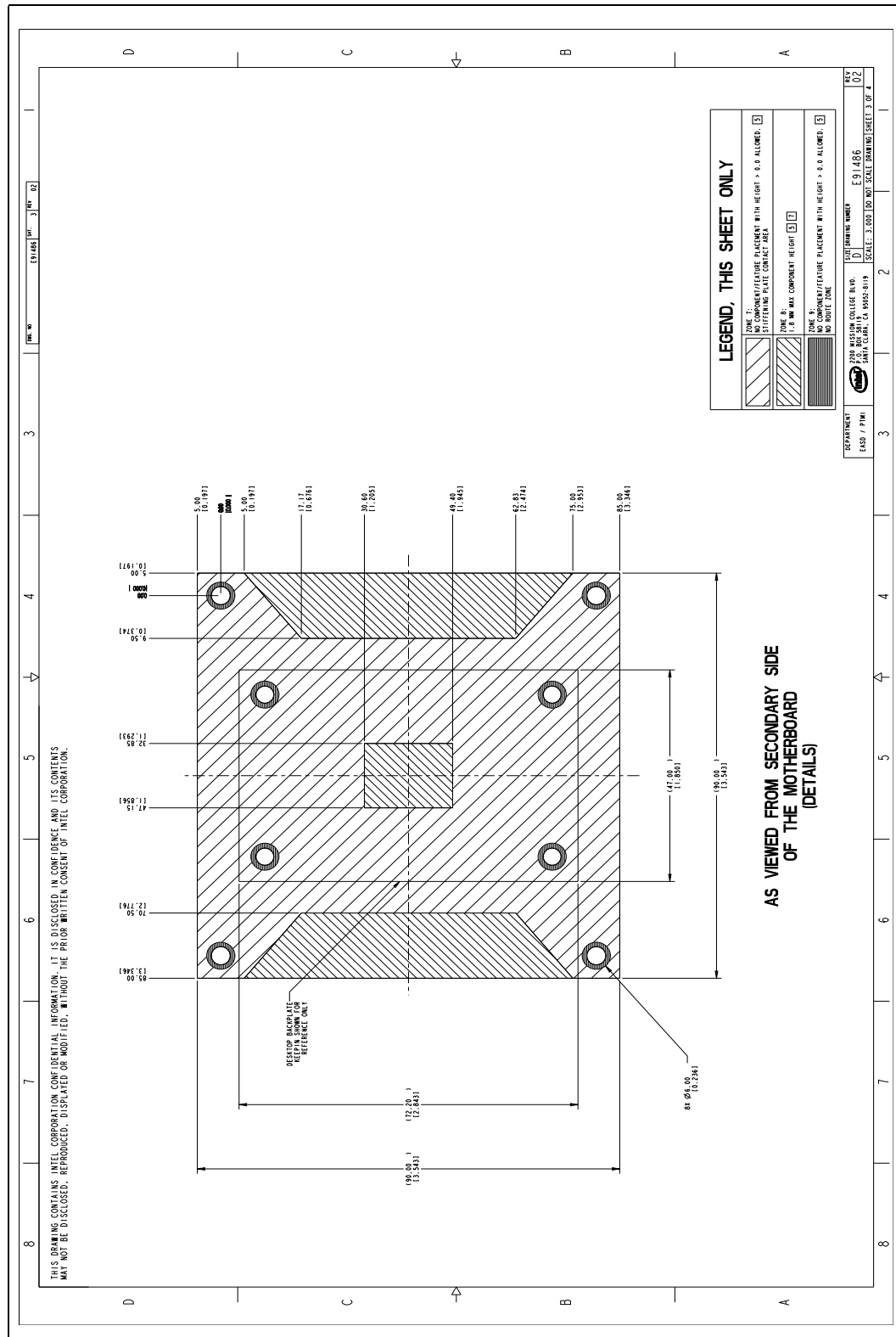
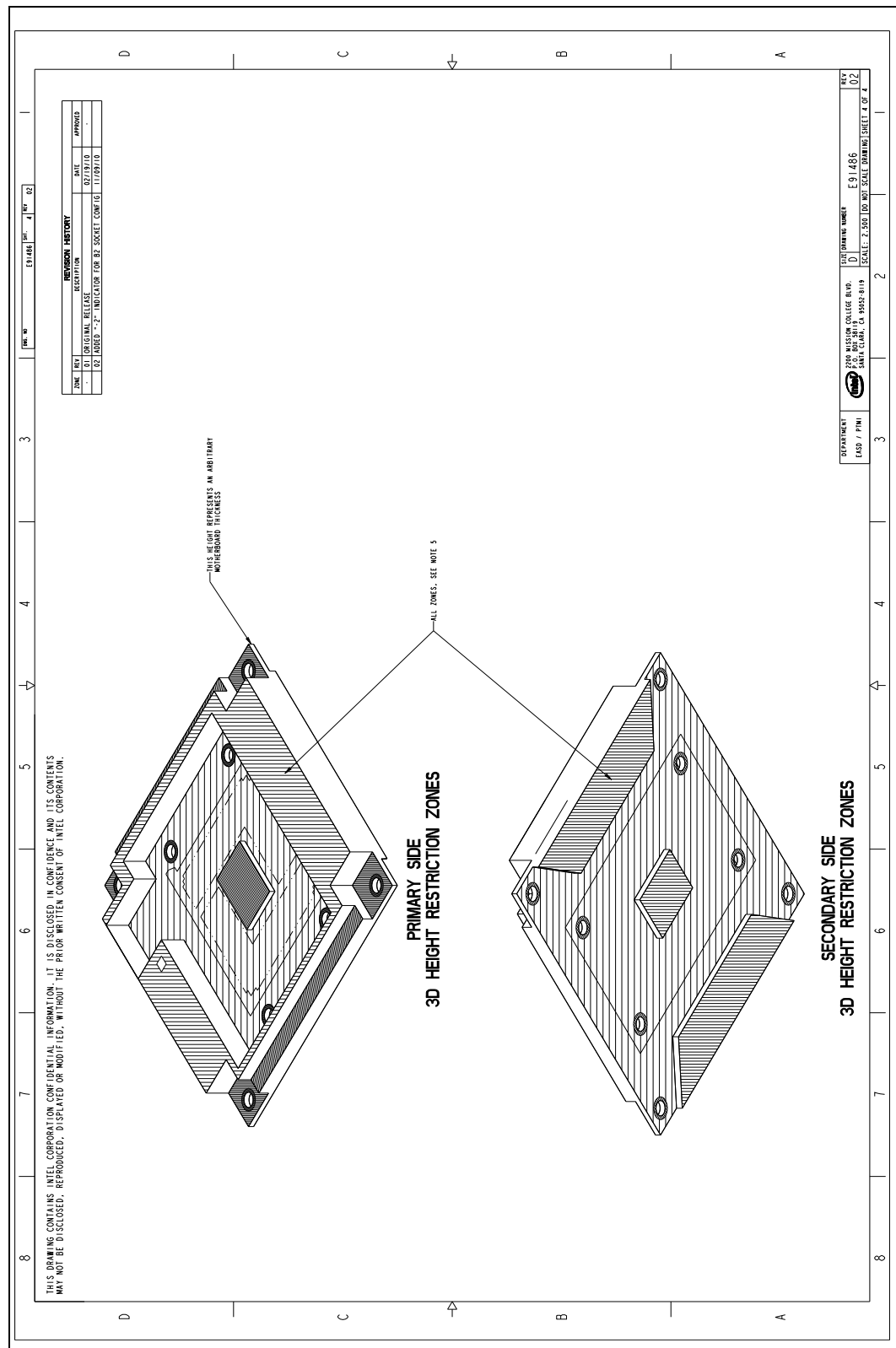
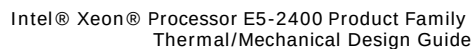


Figure B-4. Board Keepin / Keepout Zones (Sheet 4 of 4)

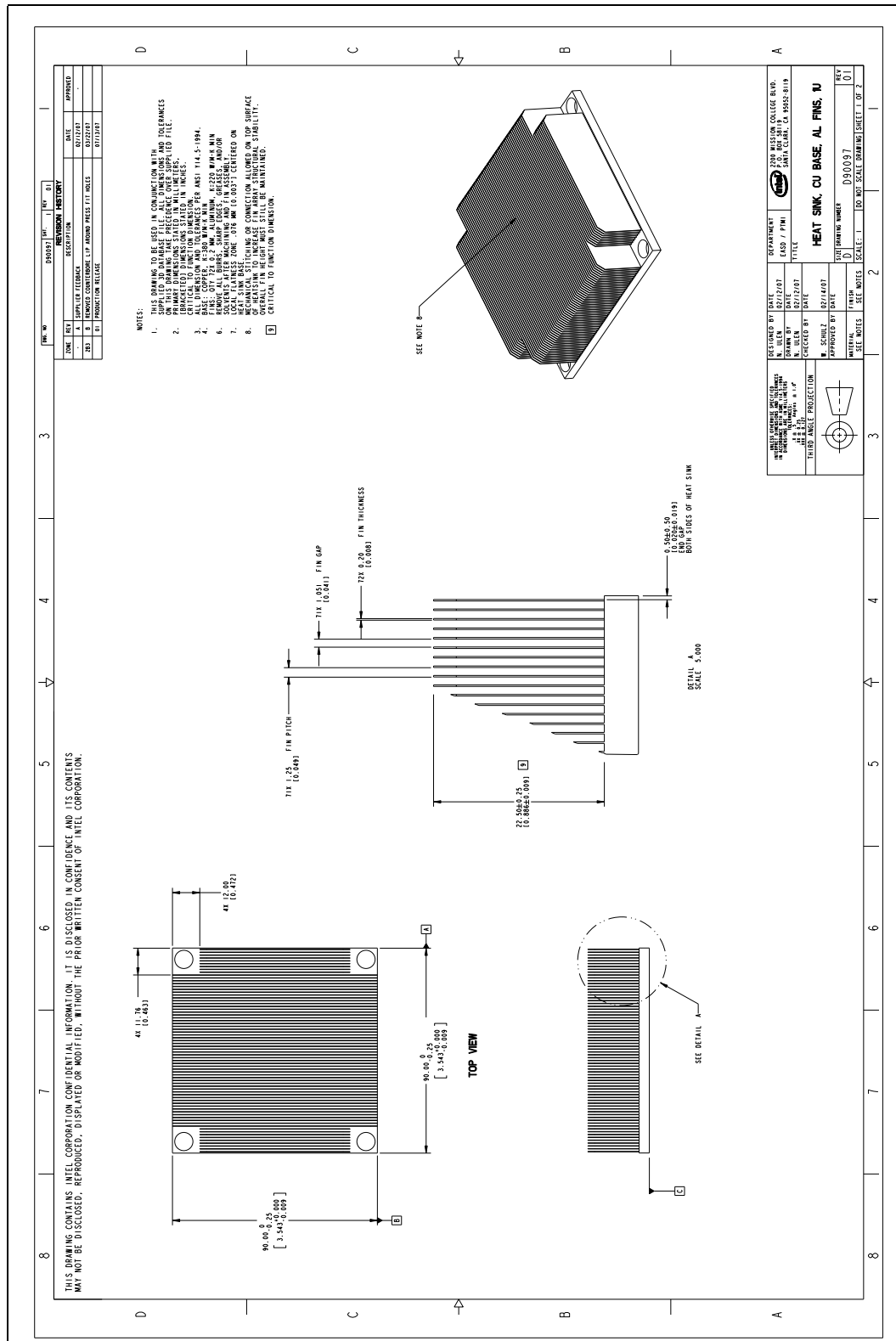


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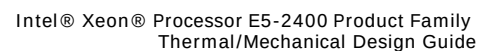
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Thermal/Mechanical Design Guide

Figure B-7. 1U Reference Heatsink Fin and Base (Sheet 1 of 2)



Intel® Xeon® Processor E5-2400 Product Family
Thermal/Mechanical Design Guide

60



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Figure B-13. 2U Collaborative Heatsink Assembly (Sheet 1 of 2)

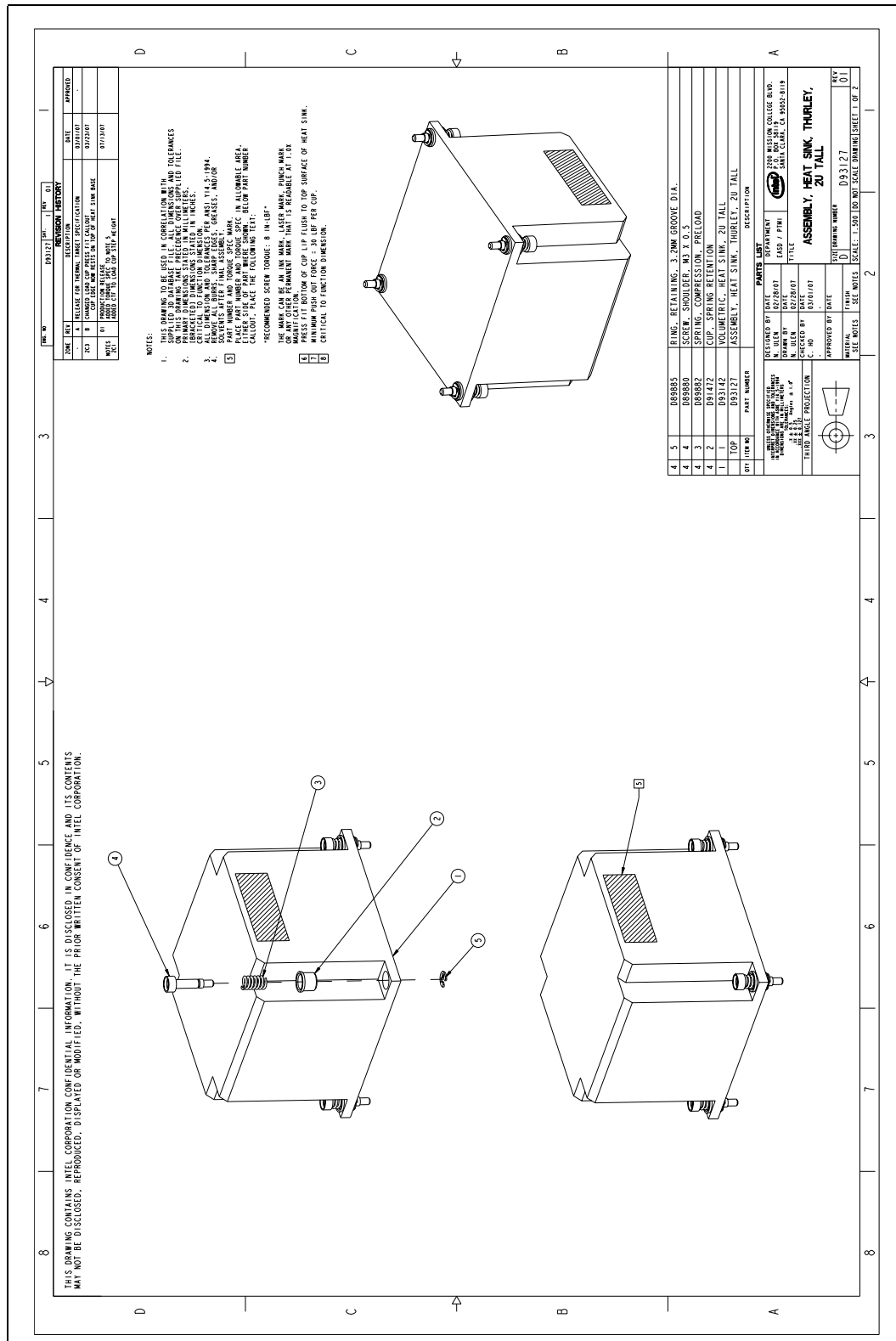
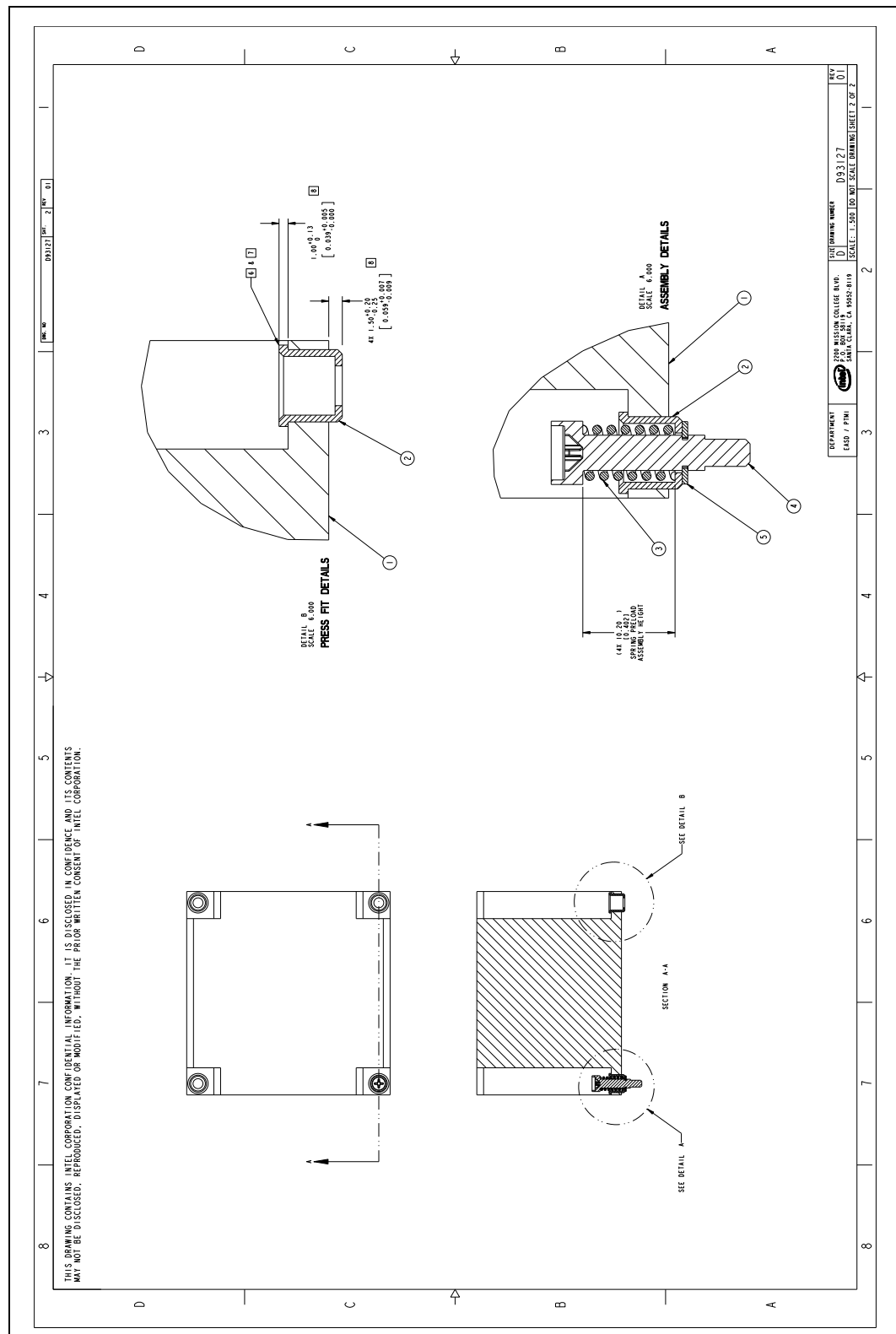


Figure B-14. 2U Collaborative Heatsink Assembly (Sheet 2 of 2)



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Figure B-17. Tower Collaborative Heatsink Assembly (Sheet 1 of 2)

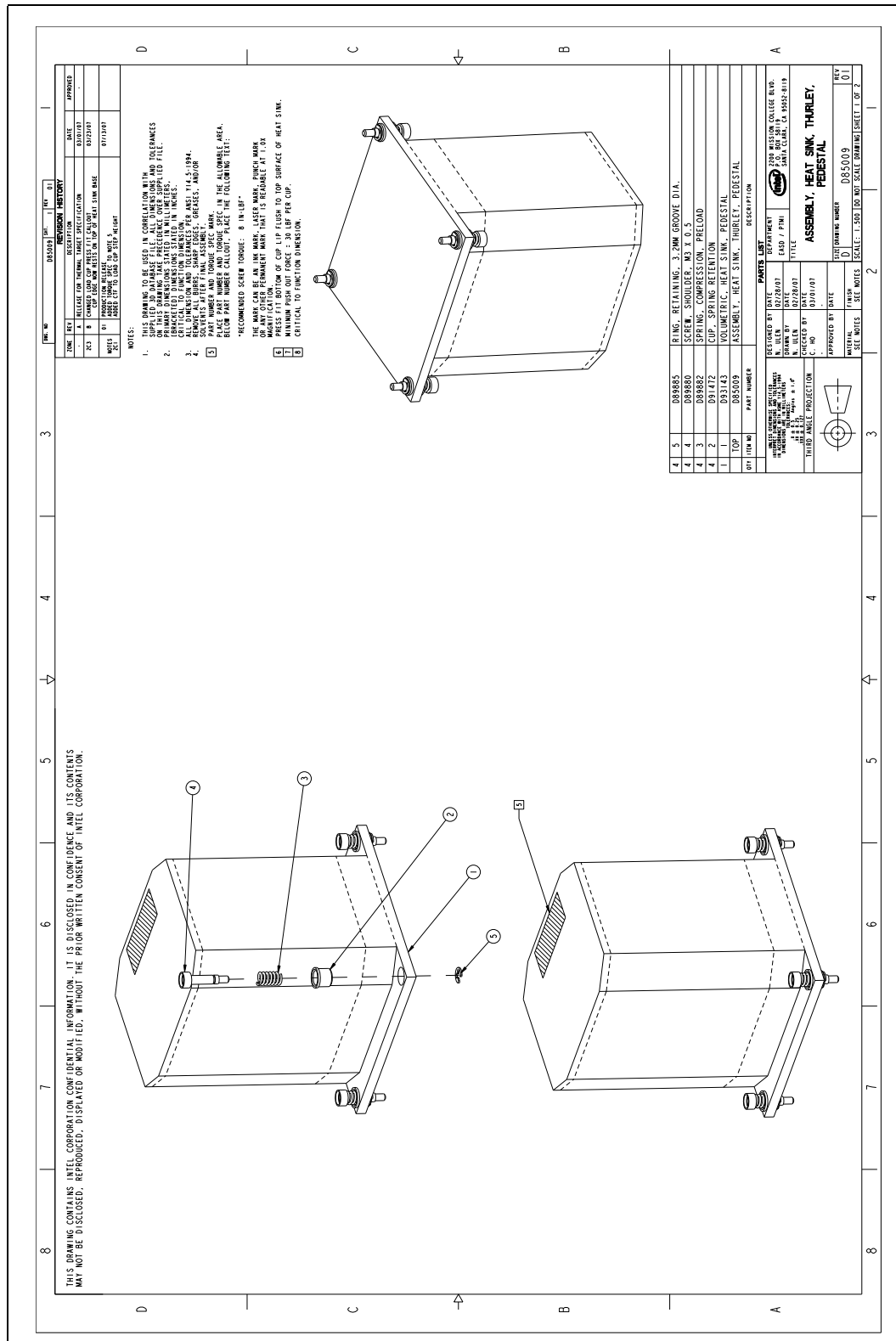
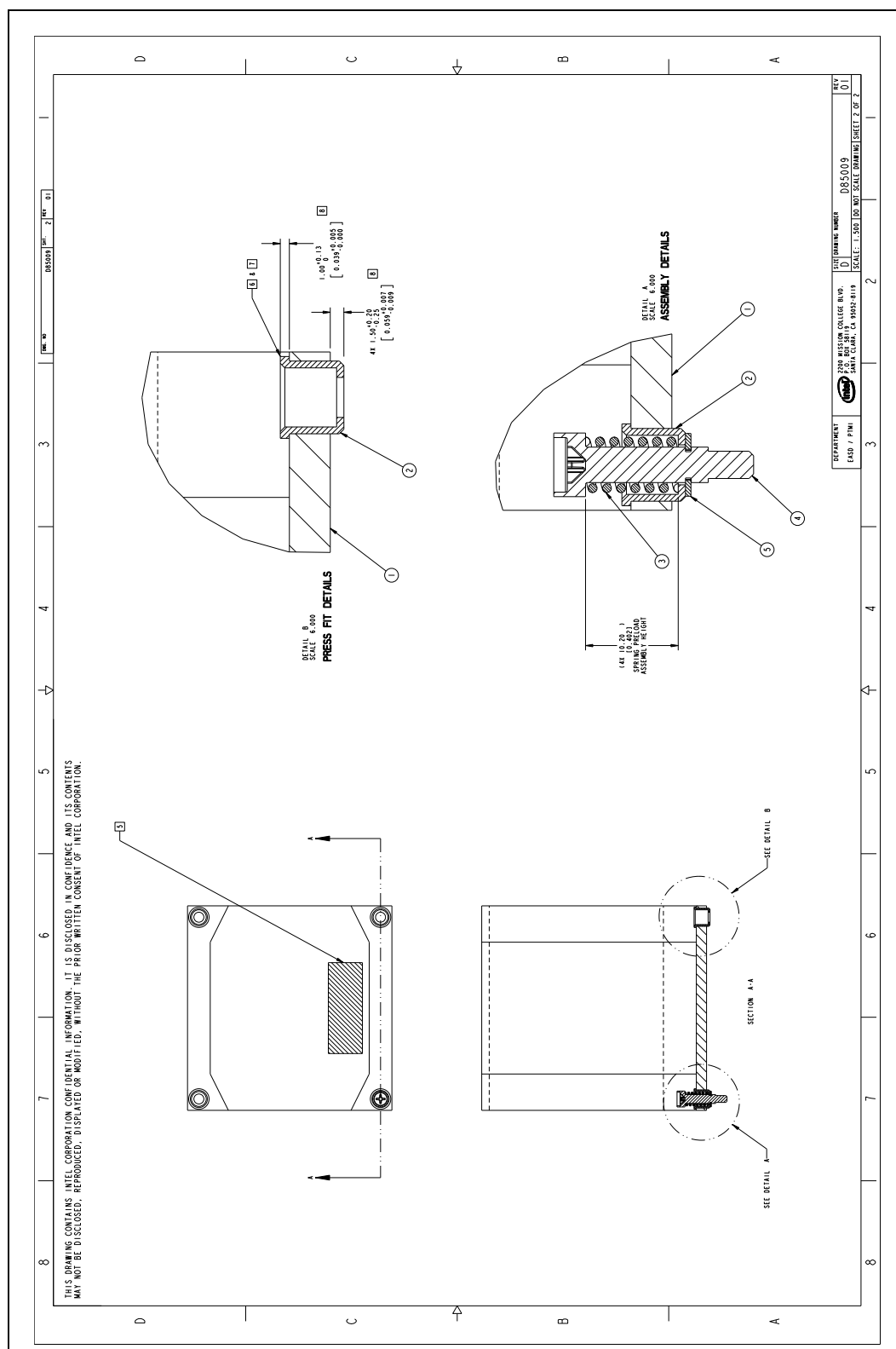
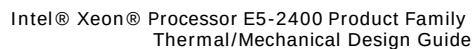


Figure B-18. Tower Collaborative Heatsink Assembly (Sheet 2 of 2)



70



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72

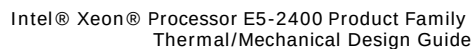
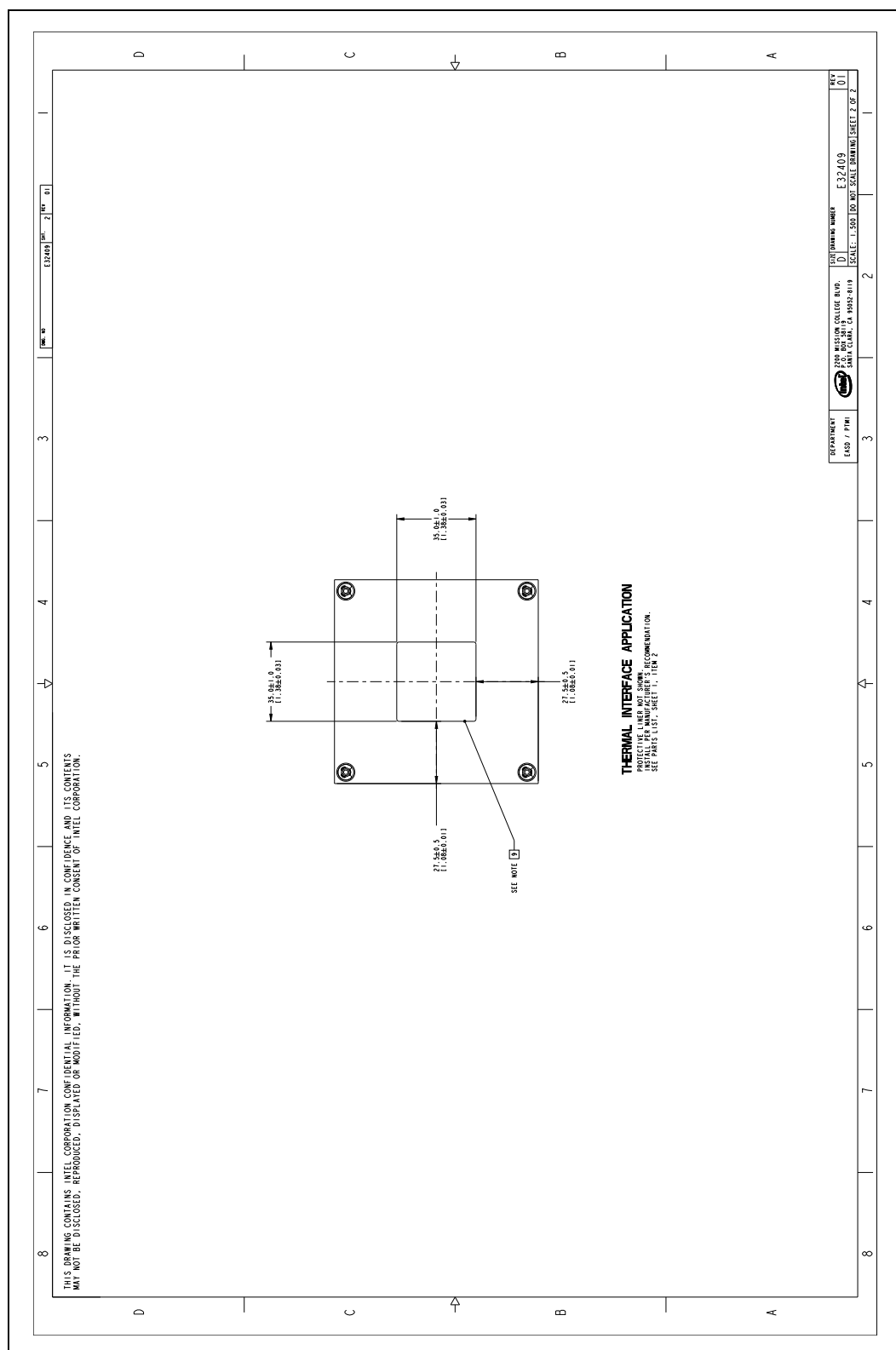
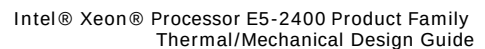


Figure B-22. 1U Reference Heatsink Assembly with TIM (Sheet 2 of 2)



74



Intel® Xeon® Processor E5-2400 Product Family
Thermal/Mechanical Design Guide

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Intel® Xeon® Processor E5-2400 Product Family
Thermal/Mechanical Design Guide

78

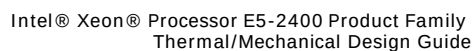
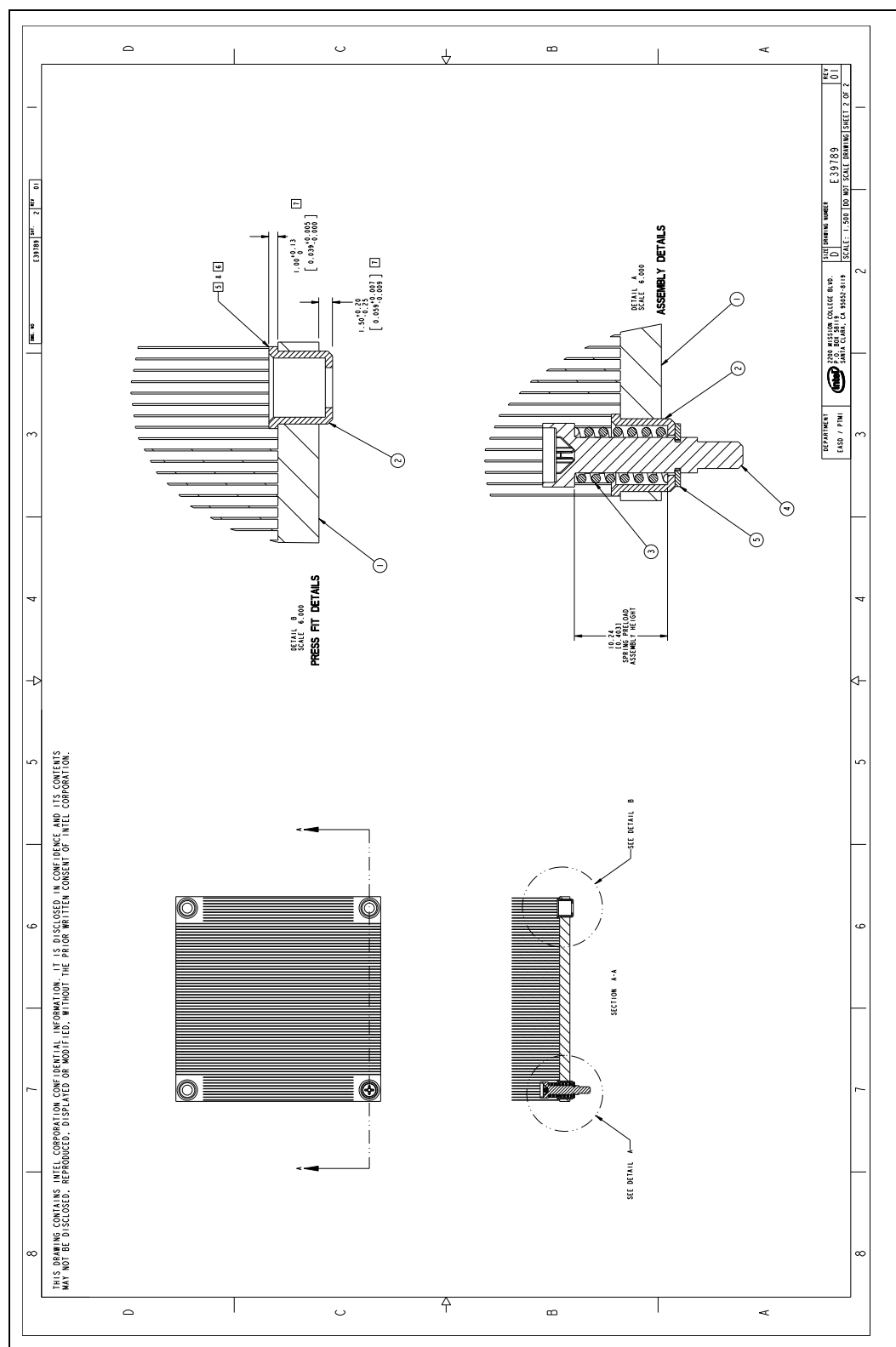


Figure B-28. 25.5 mm Reference Heatsink Assembly (Sheet 2 of 2)

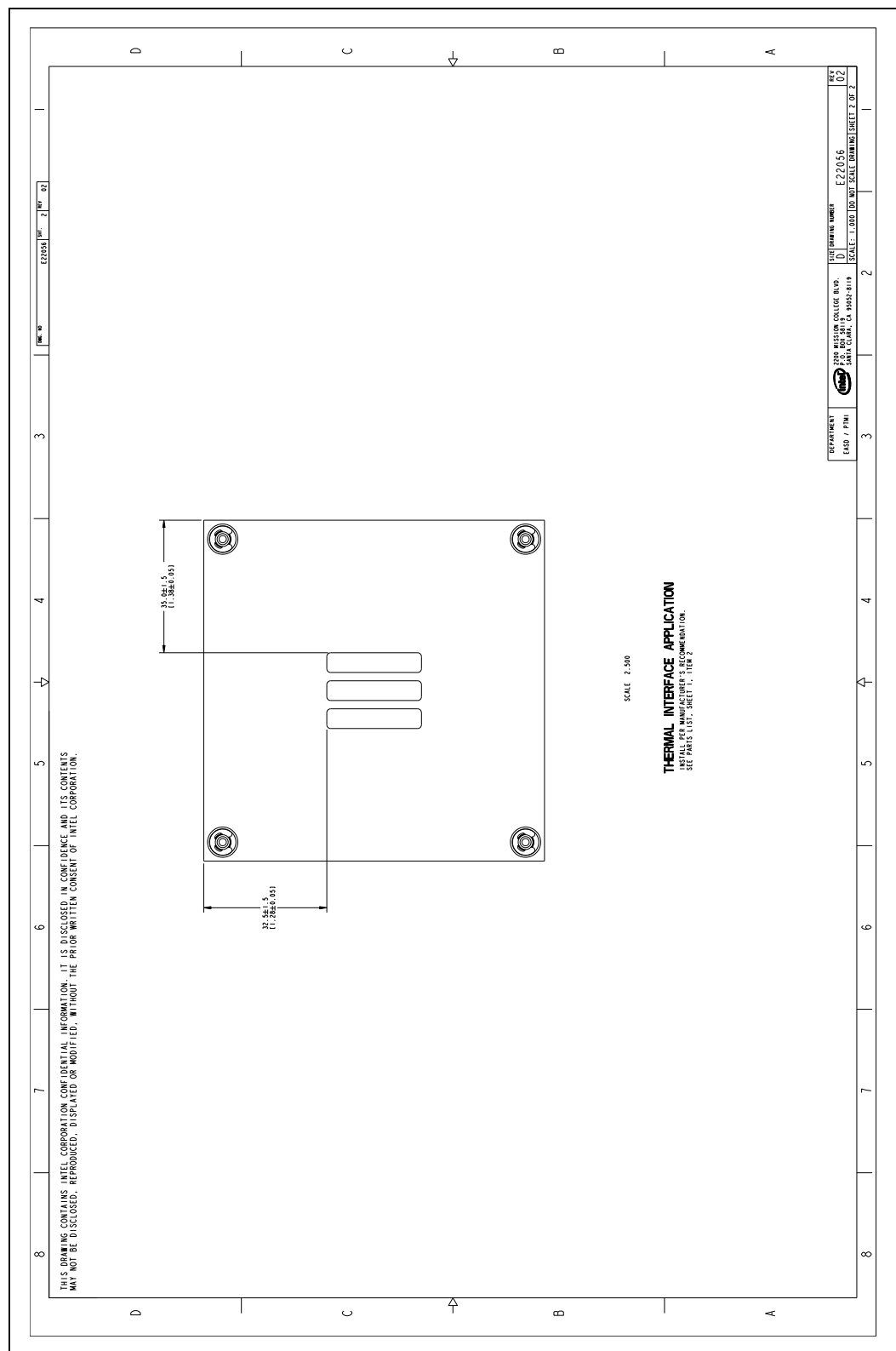


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Thermal/Mechanical Design Guide

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Figure B-32. 25.5 mm Reference Heatsink Assembly with TIM (Sheet 2 of 2)





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C Socket Mechanical Drawings

Table C-1 lists the mechanical drawings included in this appendix.

Table C-1. Mechanical Drawing List

Drawing Description	Figure Number
"Socket Mechanical Drawing (Sheet 1 of 4)"	Figure C-1
"Socket Mechanical Drawing (Sheet 2 of 4)"	Figure C-2
"Socket Mechanical Drawing (Sheet 3 of 4)"	Figure C-3
"Socket Mechanical Drawing (Sheet 4 of 4)"	Figure C-4

Figure C-1. Socket Mechanical Drawing (Sheet 1 of 4)

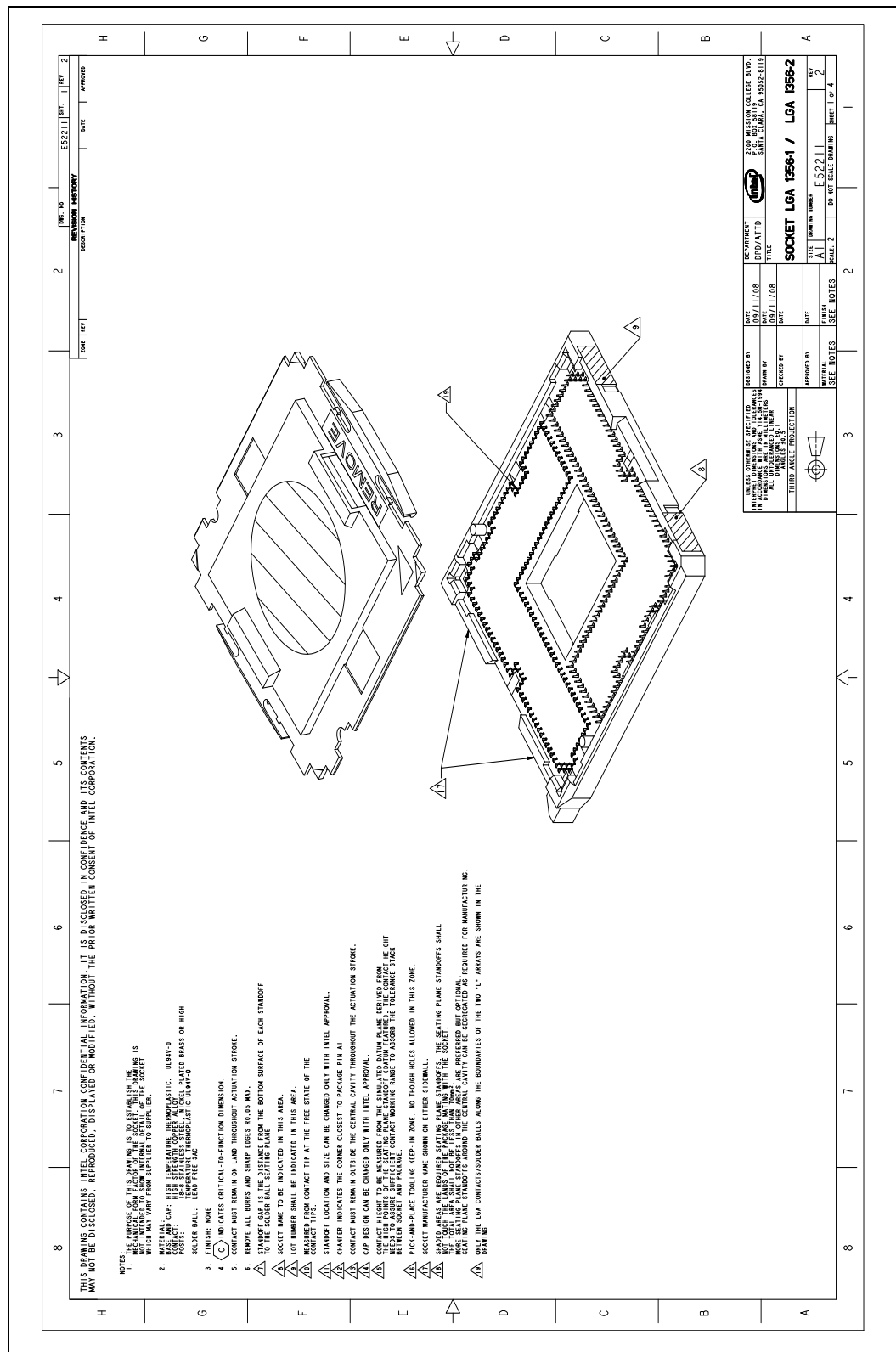
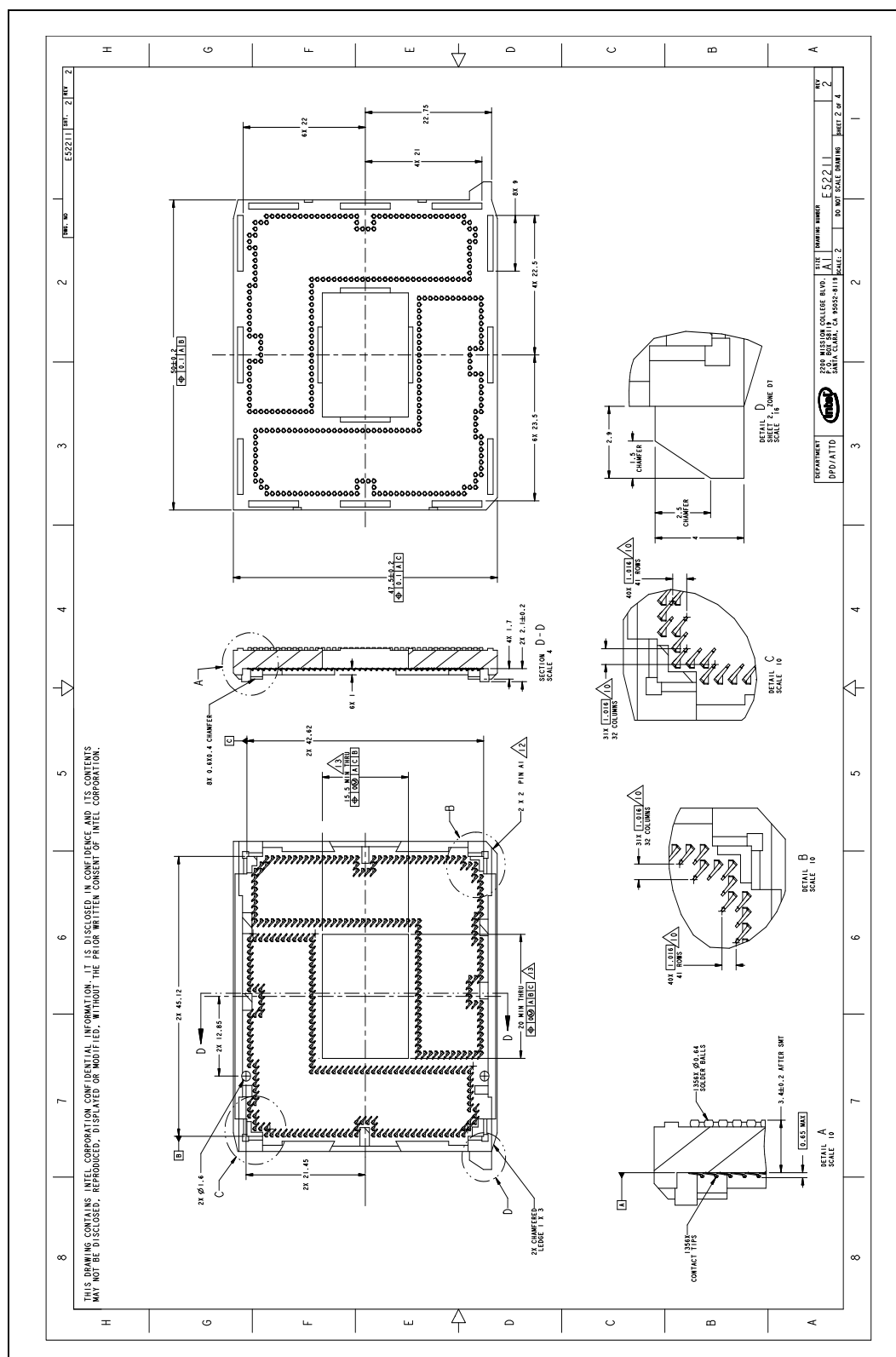
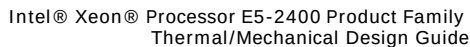


Figure C-2. Socket Mechanical Drawing (Sheet 2 of 4)



88





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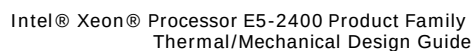
D Processor Installation Tool

The following optional tool is designed to provide mechanical assistance during processor installation and removal.

Contact the supplier for details regarding this tool:

Billy Hsieh
billy.hsieh@tycoelectronics.com
+81 44 844 8292

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E Embedded Thermal Solutions

Embedded Server SKU's target higher case temperatures and/or NEBS thermal profiles for embedded communications server and storage form factors. This section describes reference heatsinks for NEBS (Network Equipment Building Systems) compliant ATCA (Advanced Telecommunications Computing Architecture) systems. These higher case temperature processors are sufficient for any form factor that needs to meet NEBS requirements.

E.1 Performance Targets

Table E-1 provides boundary conditions and performance targets for 1U and ATCA heatsinks. These values are used to generate processor thermal specifications and to provide guidance for heatsink design.

Table E-1. 8-Core / 6-Core Processor Reference Thermal Boundary Conditions

TDP	Heatsink Technology ^{5, 6}	Ψ_{ca2} (°C/W)	$T_{LA}^{1, 4}$ (°C)	Heatsink Volumetric ³ (mm)
LV70W (8-core)	ATCA Cu base Al fins	0.466	45/60	90x90x13.3
LV60W (6-core)	ATCA Cu base Al fins	0.467	45/60	90x90x13.3

Table E-2. 4-Core Processor Reference Thermal Boundary Conditions

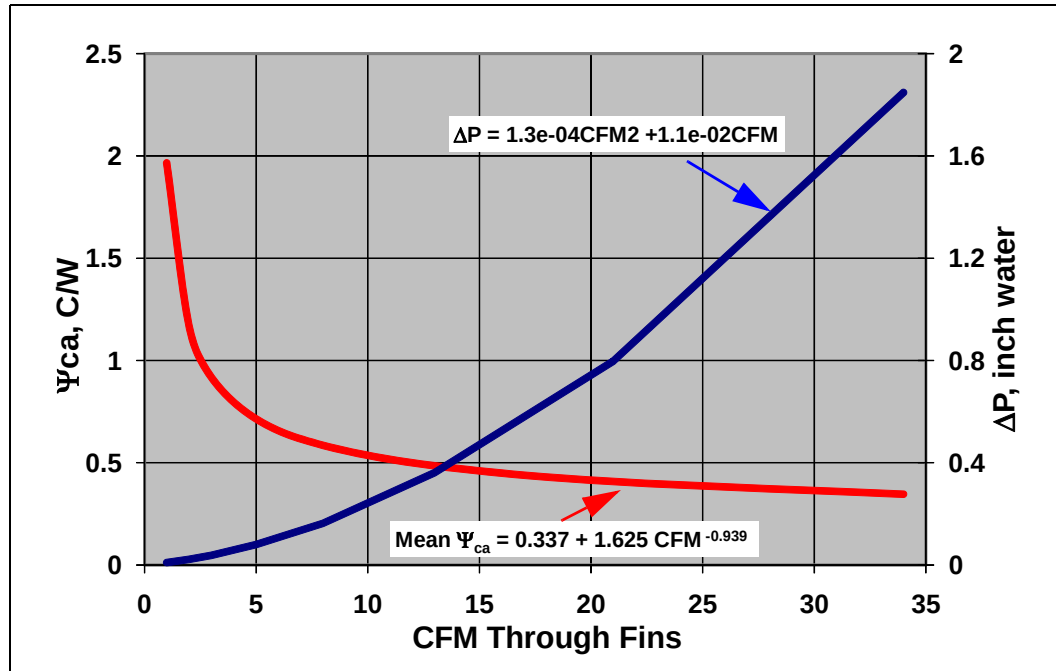
TDP	Heatsink Technology ^{5, 6}	Ψ_{ca2} (°C/W)	$T_{LA}^{1, 4}$ (°C)	Heatsink Volumetric ³ (mm)
LV50W (4-core)	ATCA Cu base Al fins	0.509	52/67	90x90x13.3

Notes:

1. Local ambient temperature of the air entering the heatsink.
2. Max target (mean + 3 sigma + offset) for thermal characterization parameter (Section 5.4.1).
3. Dimensions of heatsink do not include socket or processor.
4. Local Ambient Temperature written X/Y° C means X° C under Nominal conditions but Y° C is allowed for Short-Term NEBS excursions.
5. All heatsinks are Non-Direct Chassis Attach (DCA)
6. See Section 5.1 for standard 1U solutions that do not need to meet NEBS.

Detailed drawings for the ATCA reference heatsink can be found in [Section E.3](#). [Table E-1](#) above specifies Ψ_{ca} and pressure drop targets and [Figure E-1](#) below shows Ψ_{ca} and pressure drop for the ATCA heatsink versus the airflow provided. Best-fit equations are provided to prevent errors associated with reading the graph.

Figure E-1. ATCA Heatsink Performance Curves



Other LGA1366 compatible thermal solutions may work with the same retention.

ATCA 13 mm heatsink performance using Intel® Xeon® processor 5500 series TTV.

E.2 Thermal Design Guidelines

E.2.1 High Case Temperature Thermal Profile

Processors that offer a High case temperature thermal profile are specified in the *Intel® Xeon® Processor E5-2400 Product Family Datasheet - Volume One*.

High case temperature thermal profiles help relieve thermal constraints for Short-Term NEBS conditions. To help reliability, processors must meet the nominal thermal profile under standard operating conditions and can only rise up to the Short-Term spec for NEBS excursions (see [Figure E-2](#)). The definition of Short-Term time is clearly defined for NEBS Level 3 conditions but the key is that it cannot be longer than 360 hours per year.

Fan speed control is treated the same as standard processors. When DTS (Digital Temperature Sensor) value is less than Tcontrol, the thermal profile can be ignored.

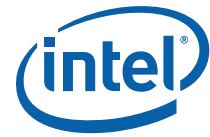
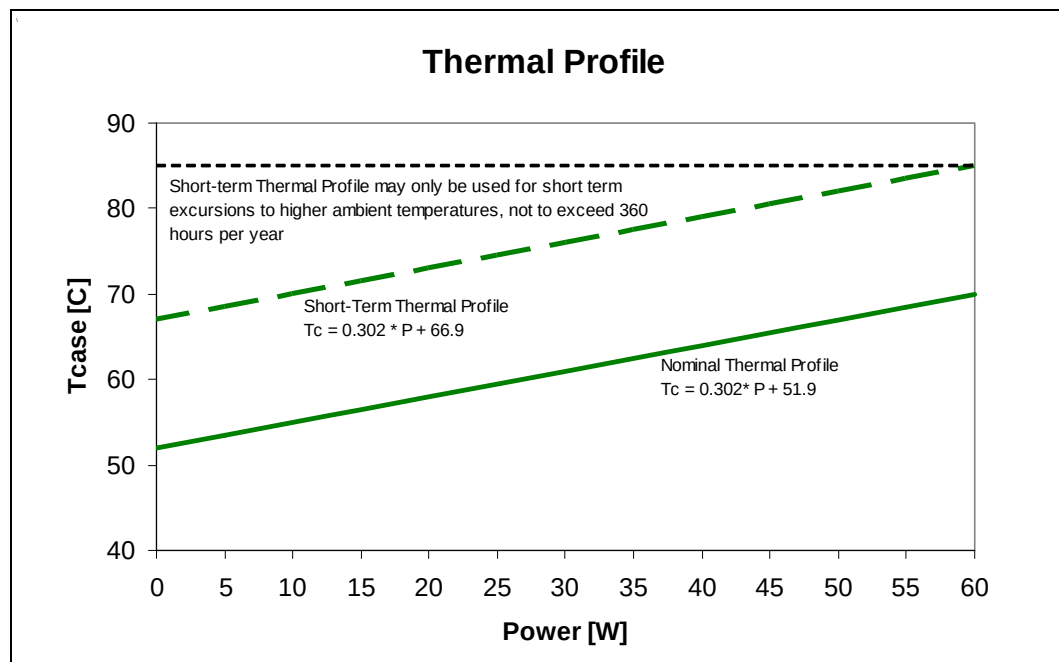


Figure E-2. NEBS Thermal Profile



Notes:

- 1.) The Nominal Thermal Profile must be used for all normal operating conditions, or for products that do not require NEBS Level 3 compliance.
- 2.) The Short-Term Thermal Profile may only be used for short-term excursions to higher ambient operating temperatures, not to exceed 360 hours per year as compliant with NEBS Level 3.
- 3.) Implementation of either thermal profile should result in virtually no TCC activation.
- 4.) Utilization of a thermal solution that exceeds the Short-Term Thermal Profile, or which operates at the Short-Term Thermal Profile for a duration longer than the limits specified in Note 3 above, do not meet the processor thermal specifications and may result in permanent damage to the processor.

E.3 Mechanical Drawings and Supplier Information

See [Appendix B](#) for retention and keep out drawings.

The part number below represent Intel reference designs for a ATCA reference heatsink. Customer implementation of these components may be unique and require validation by the customer. Customers can obtain these components directly from the supplier below.

Table E-3. Embedded Heatsink Component Suppliers

Component	Description	Supplier PN	Supplier Contact Info
ATCA Reference Heatsink Intel P/N E65918-001	ATCA Copper Fin, Copper Base	Fujikura HSA-7901-B	Fujikura America Yuji Yasuda yuji@fujikura.com 408-988-7478 Fujikura Taiwan Branch Yao-Hsien Huang yeohsien@fujikuratw.com.tw 886(2)8788-4959

**Table E-4. Mechanical Drawings List**

Parameter	Value
ATCA Reference Heat Sink Assembly (Sheet 1 of 2)	Figure E-3
ATCA Reference Heat Sink Assembly (Sheet 2 of 2)	Figure E-4
ATCA Reference Heatsink Fin and Base (Sheet 1 of 2)	Figure E-5
ATCA Reference Heatsink Fin and Base (Sheet 2 of 2)	Figure E-6

Figure E-3. ATCA Reference Heat Sink Assembly (Sheet 1 of 2)

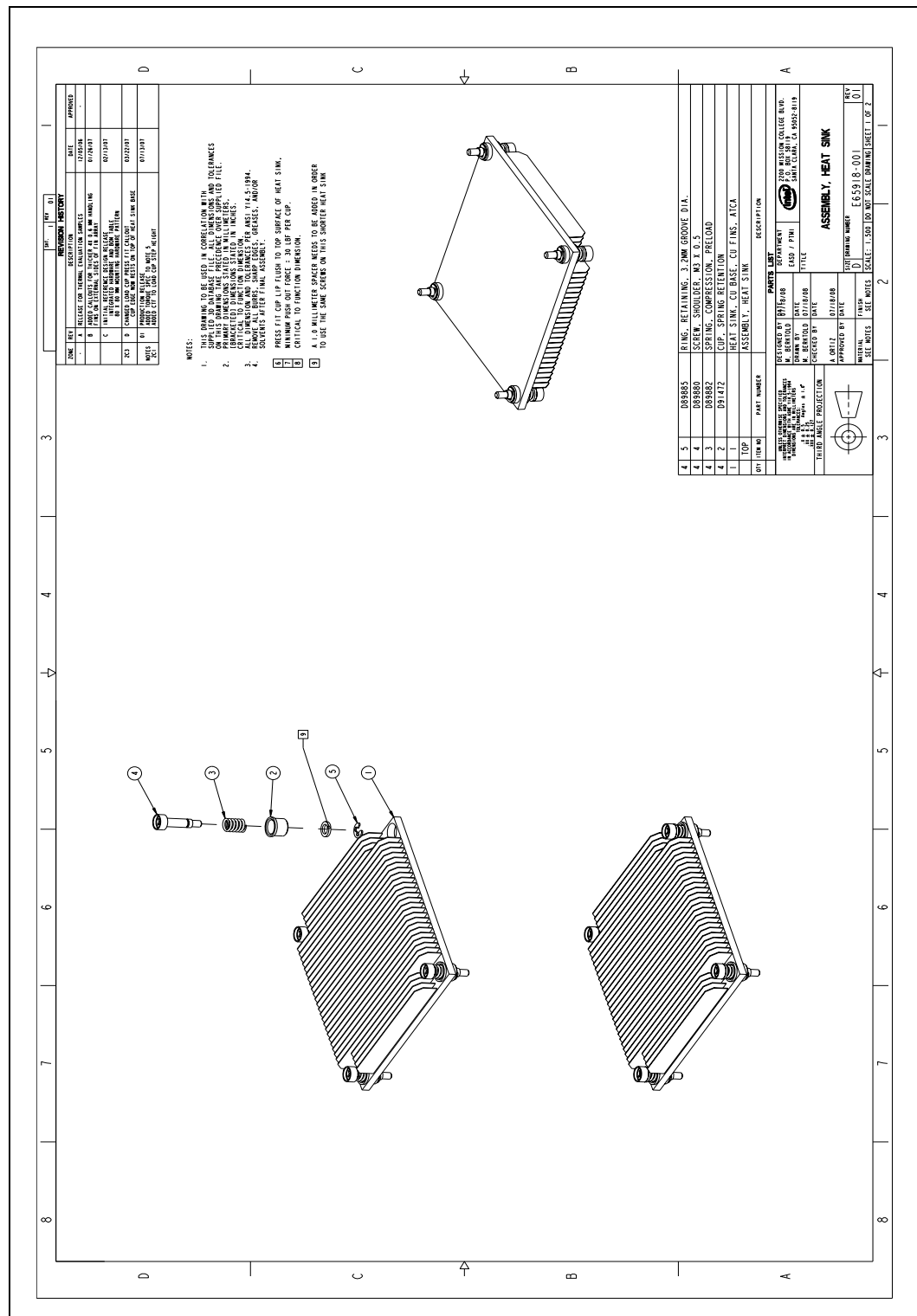
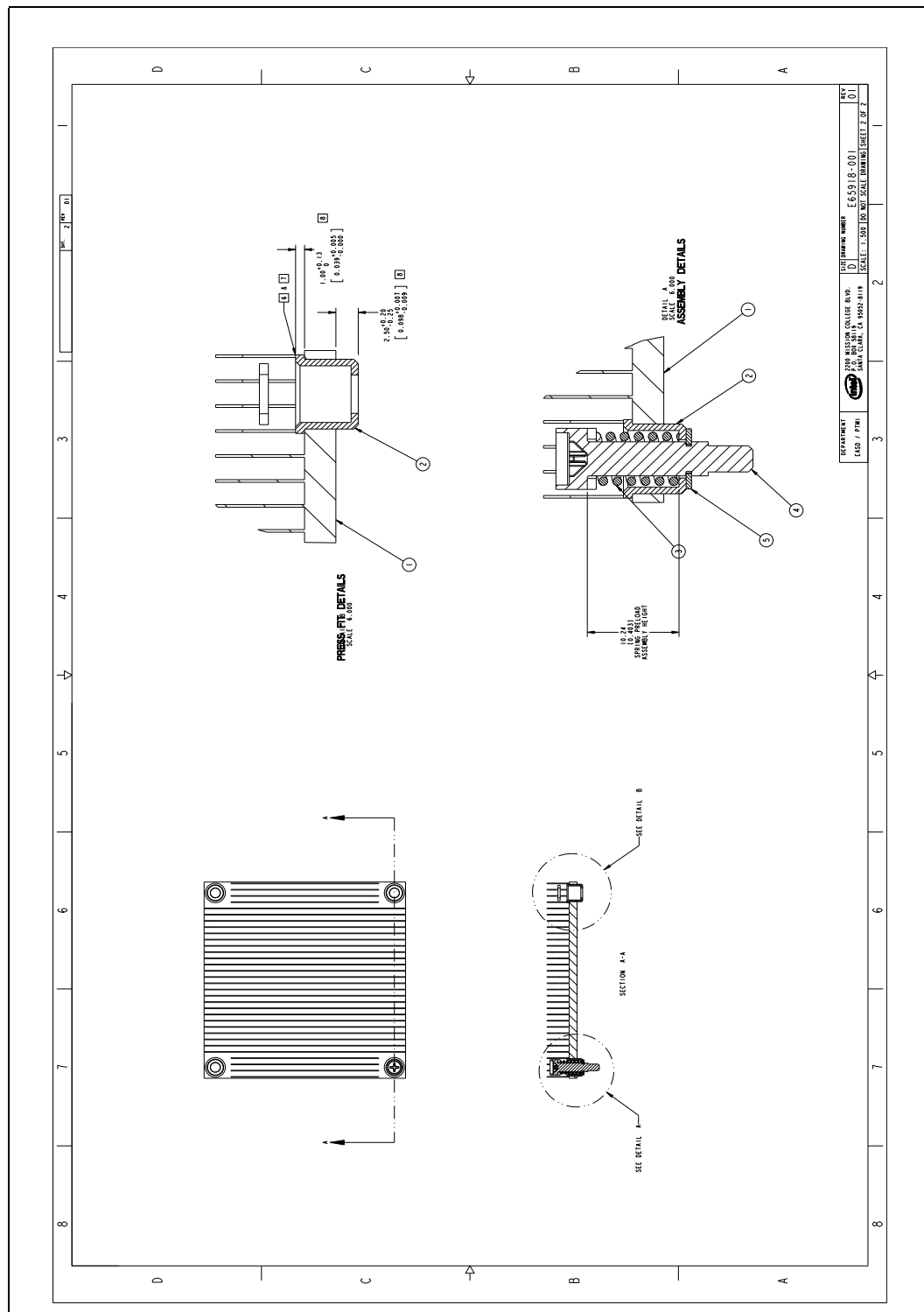


Figure E-4. ATCA Reference Heat Sink Assembly (Sheet 2 of 2)



Intel® Xeon® Processor E5-2400 Product Family
Thermal/Mechanical Design Guide

100

